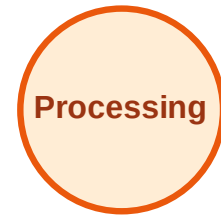


BFS Traversal

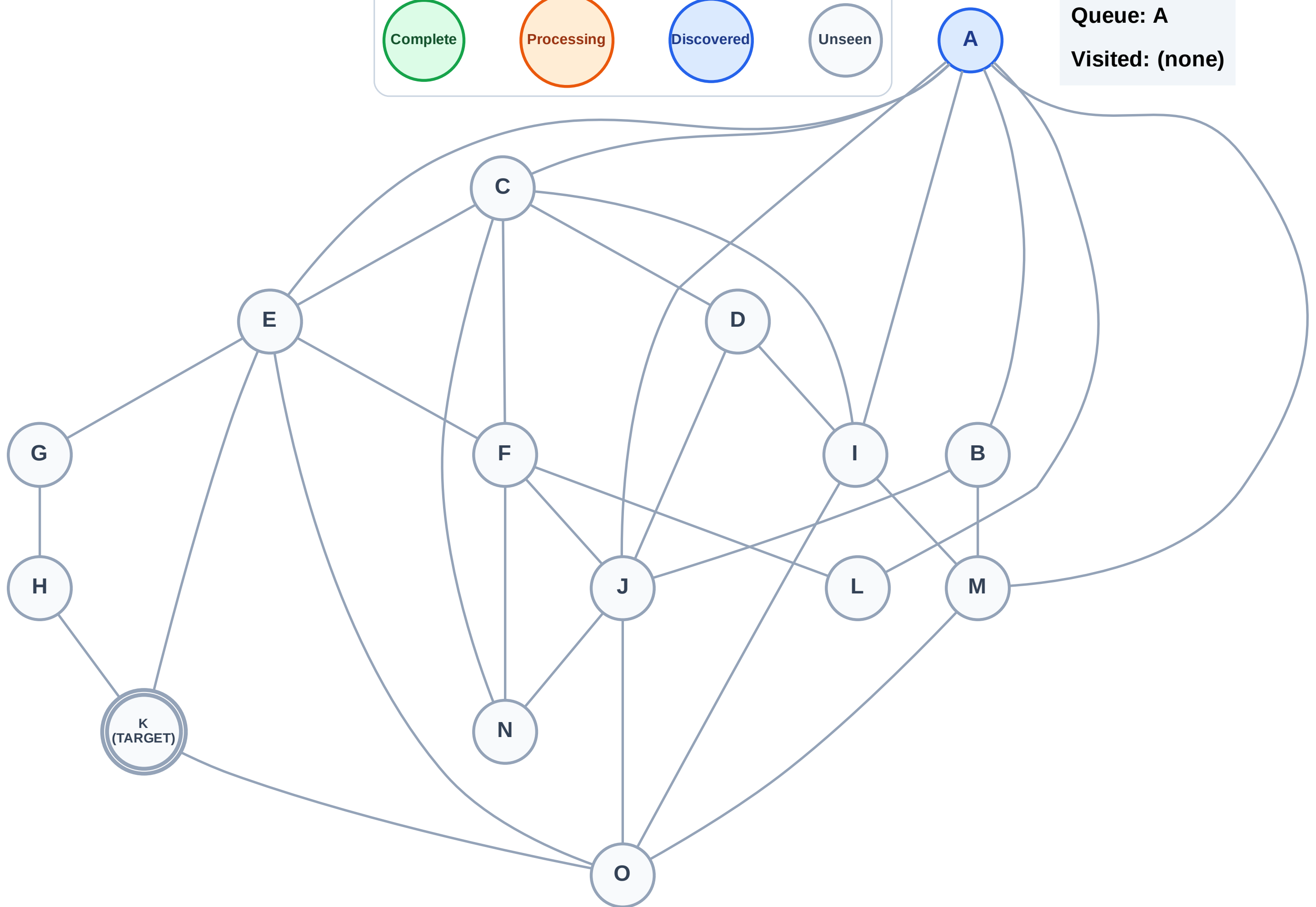
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



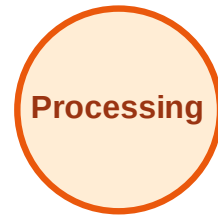
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



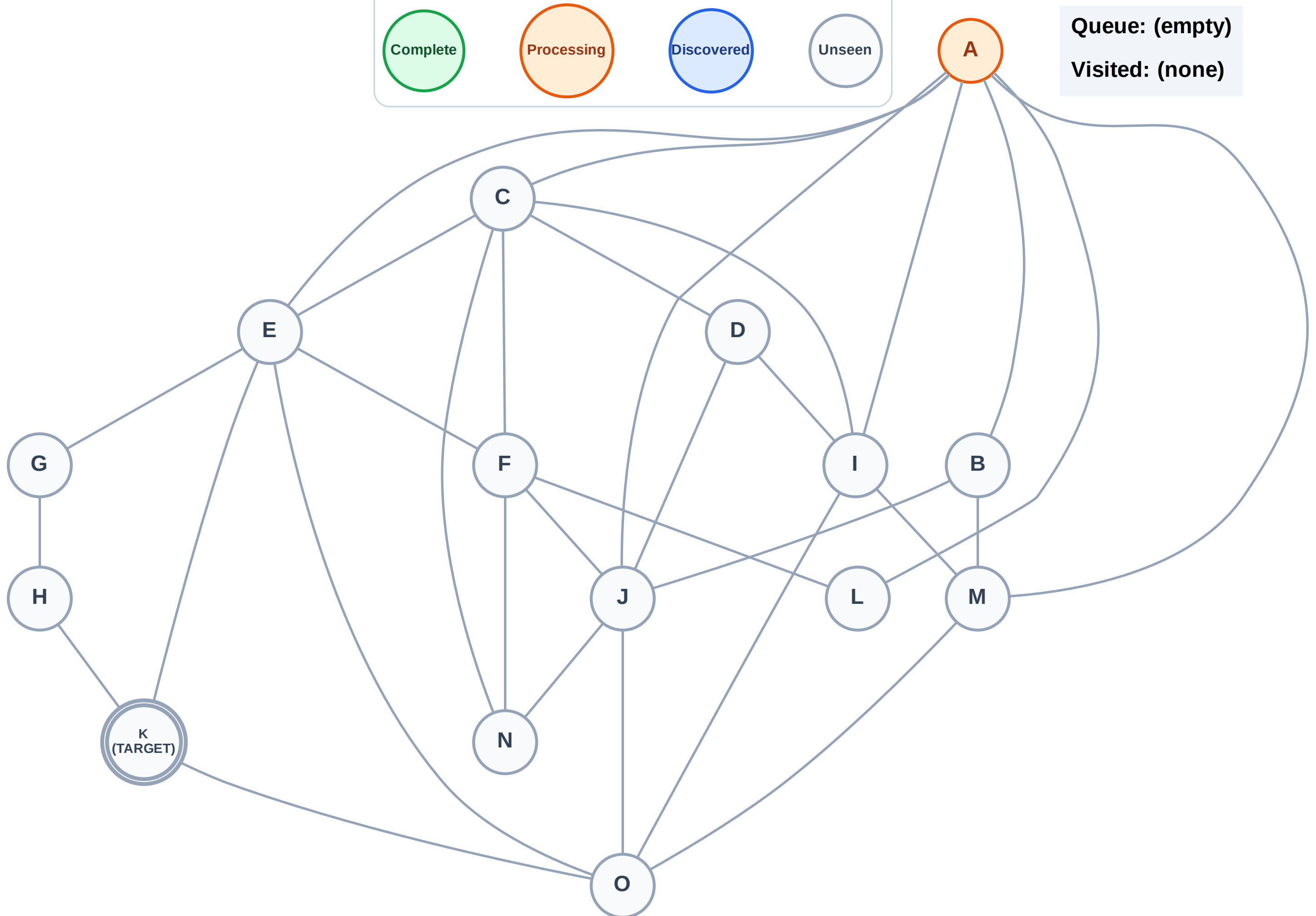
Discovered



Unseen

Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, C, E, I, J, L, M from A

Legend

Complete

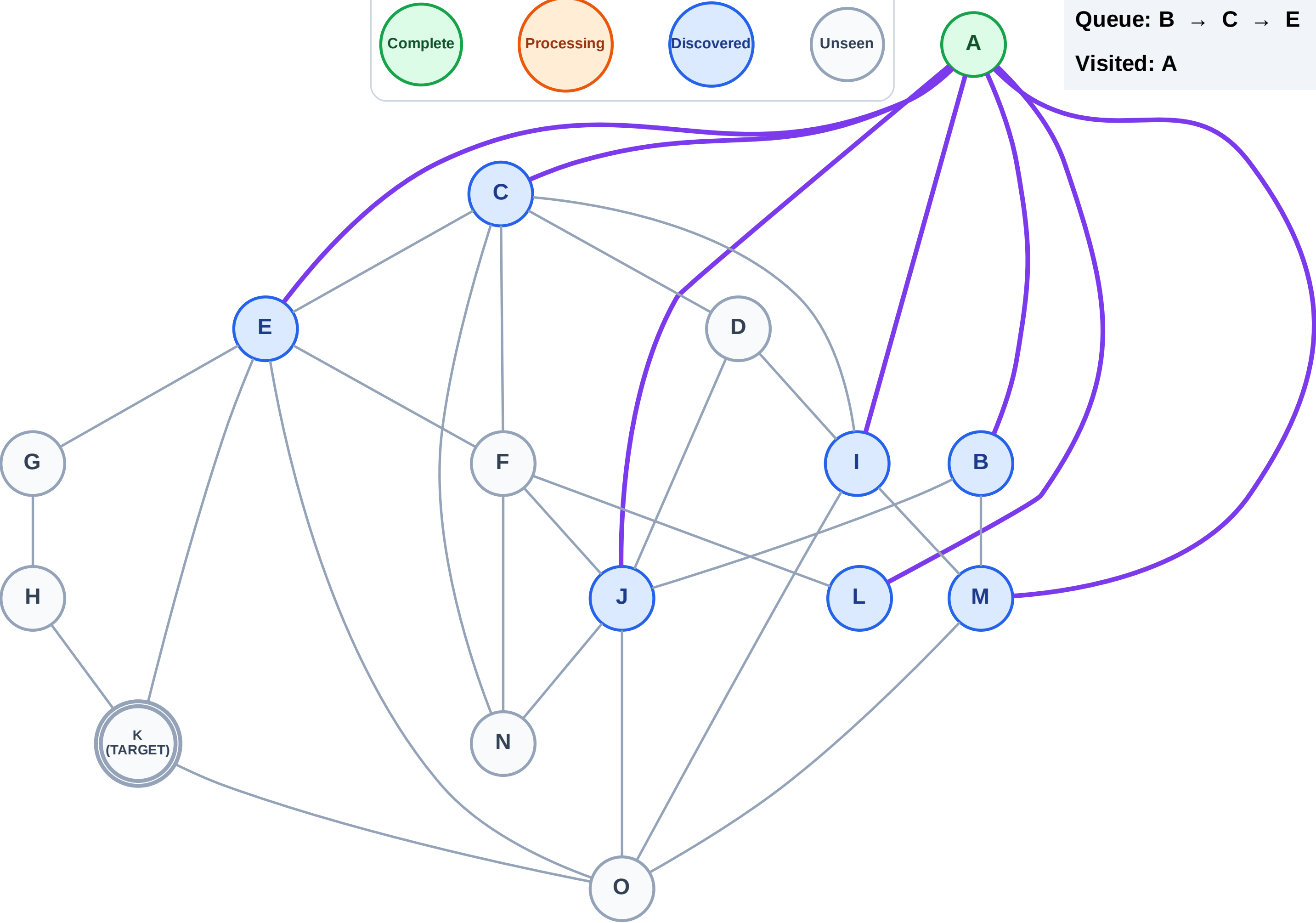
Processing

Discovered

Unseen

Queue: B → C → E → I → J → L → M

Visited: A



BFS Traversal

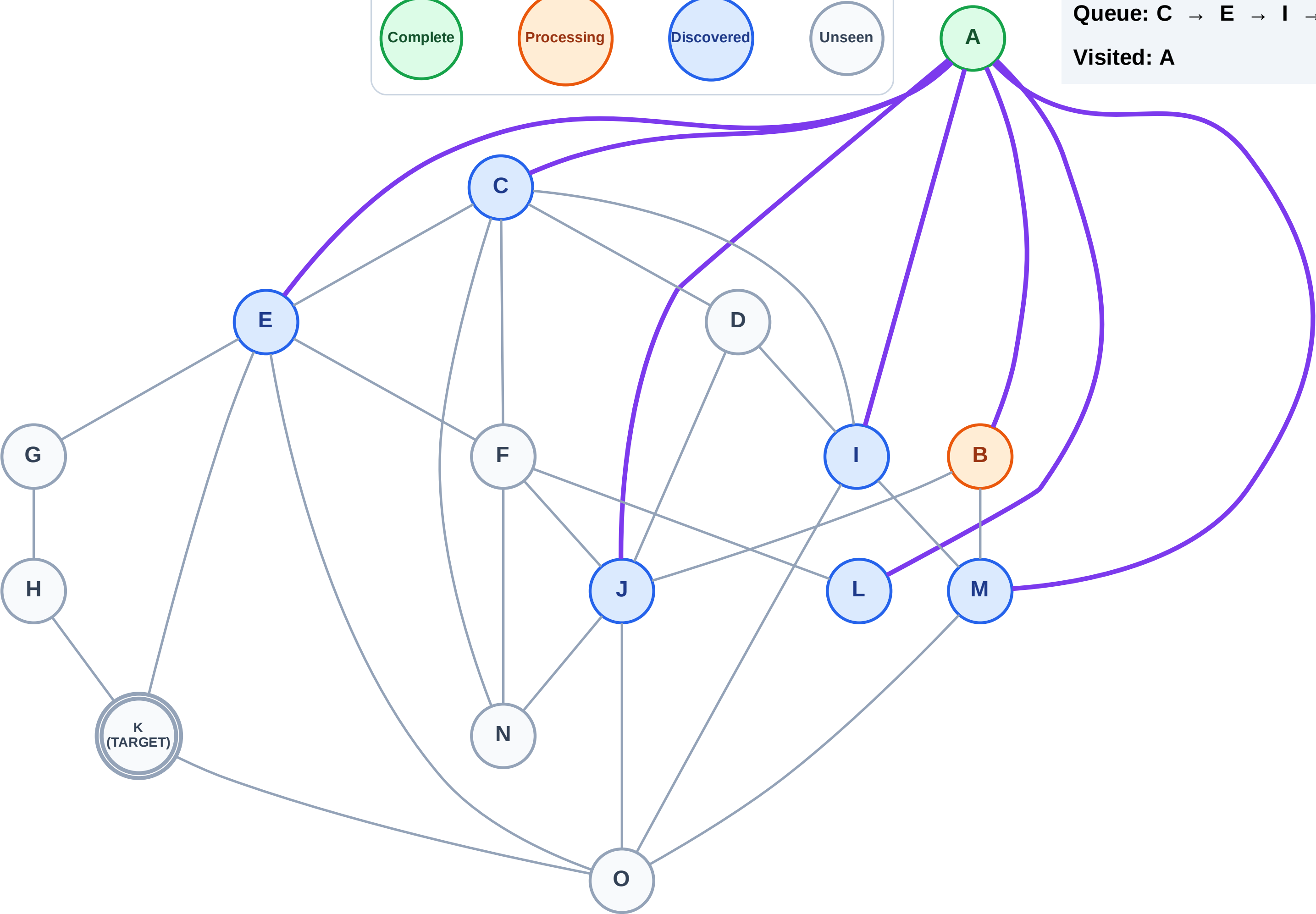
Step 4: Process node B

Legend



Queue: C → E → I → J → L → M

Visited: A



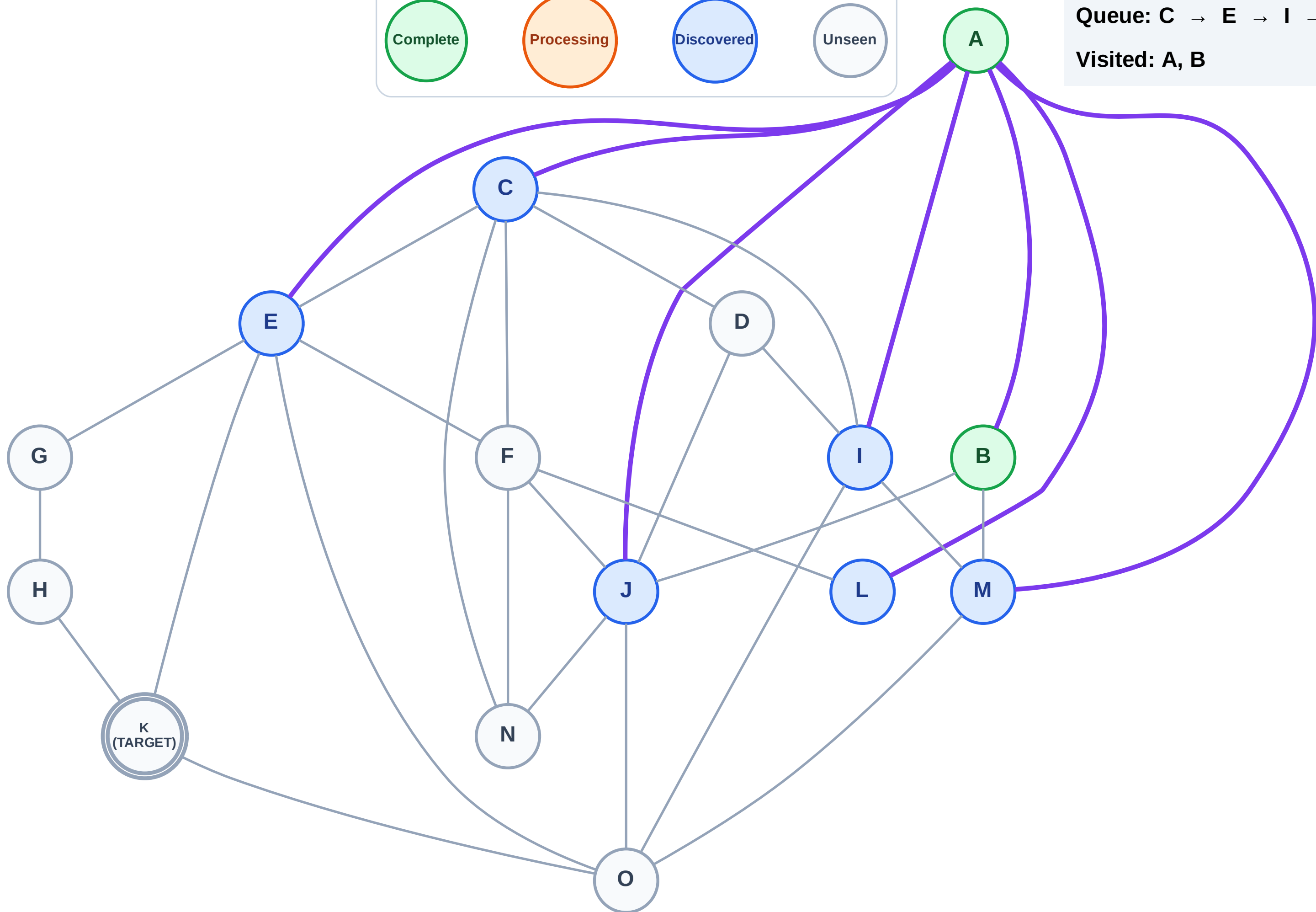
BFS Traversal

Step 5: No new neighbors from B

Legend



Queue: C → E → I → J → L → M
Visited: A, B



BFS Traversal

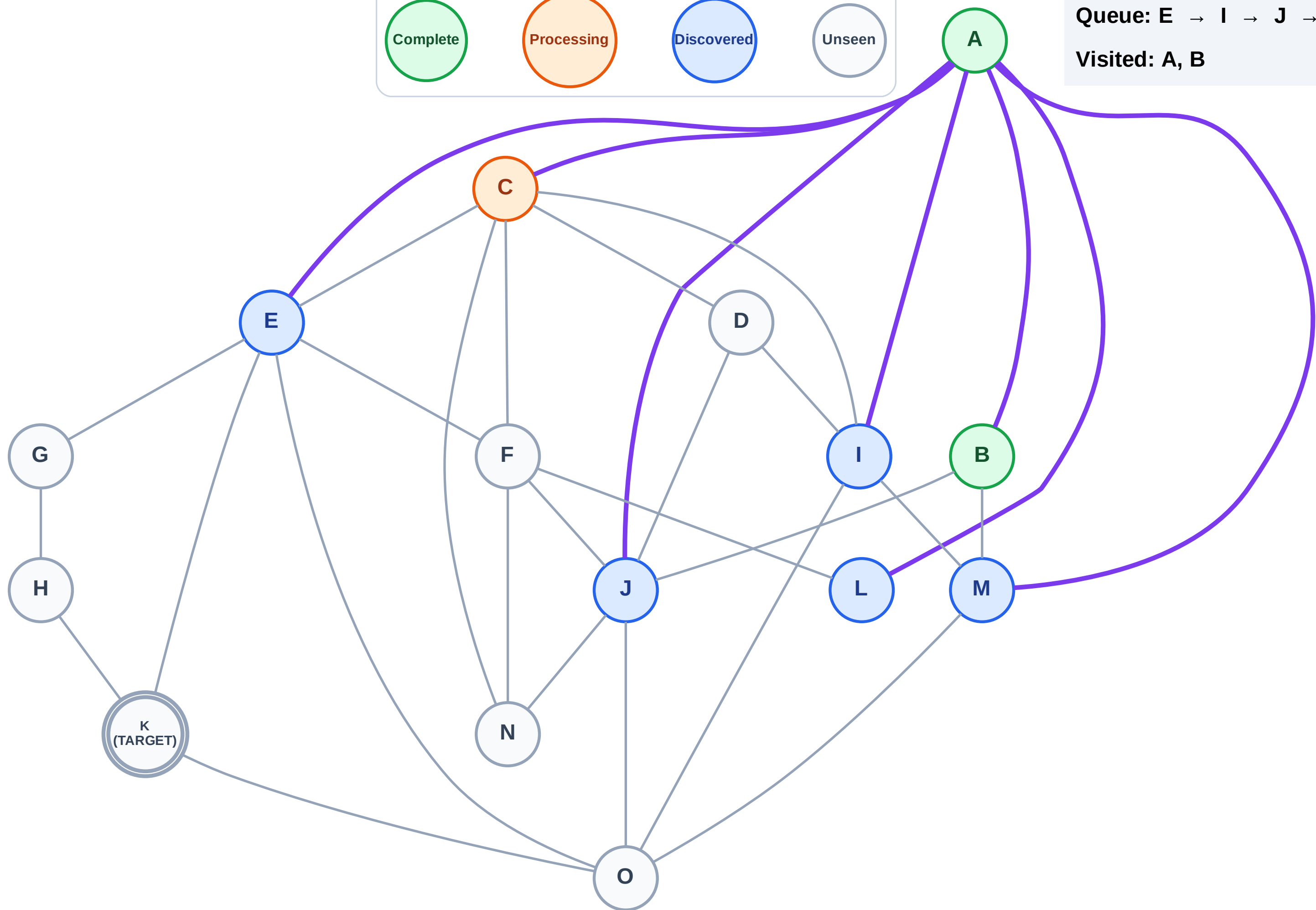
Step 6: Process node C

Legend



Queue: E → I → J → L → M

Visited: A, B



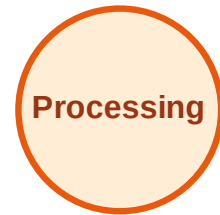
BFS Traversal

Step 7: Discover D, F, N from C

Legend



Complete



Processing



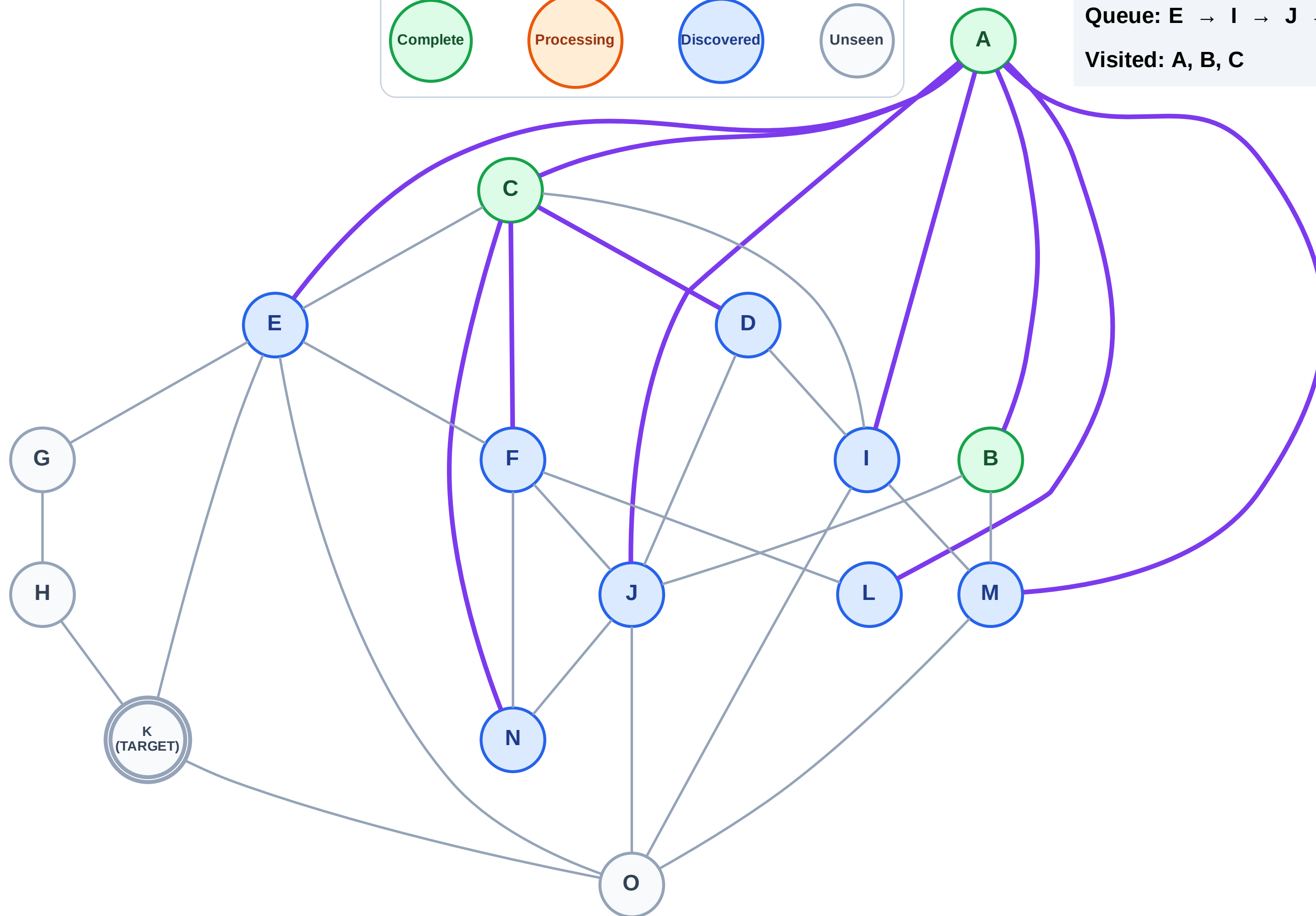
Discovered



Unseen

Queue: E → I → J → L → M → D → F → N

Visited: A, B, C



BFS Traversal

Step 8: Process node E

Legend

Complete

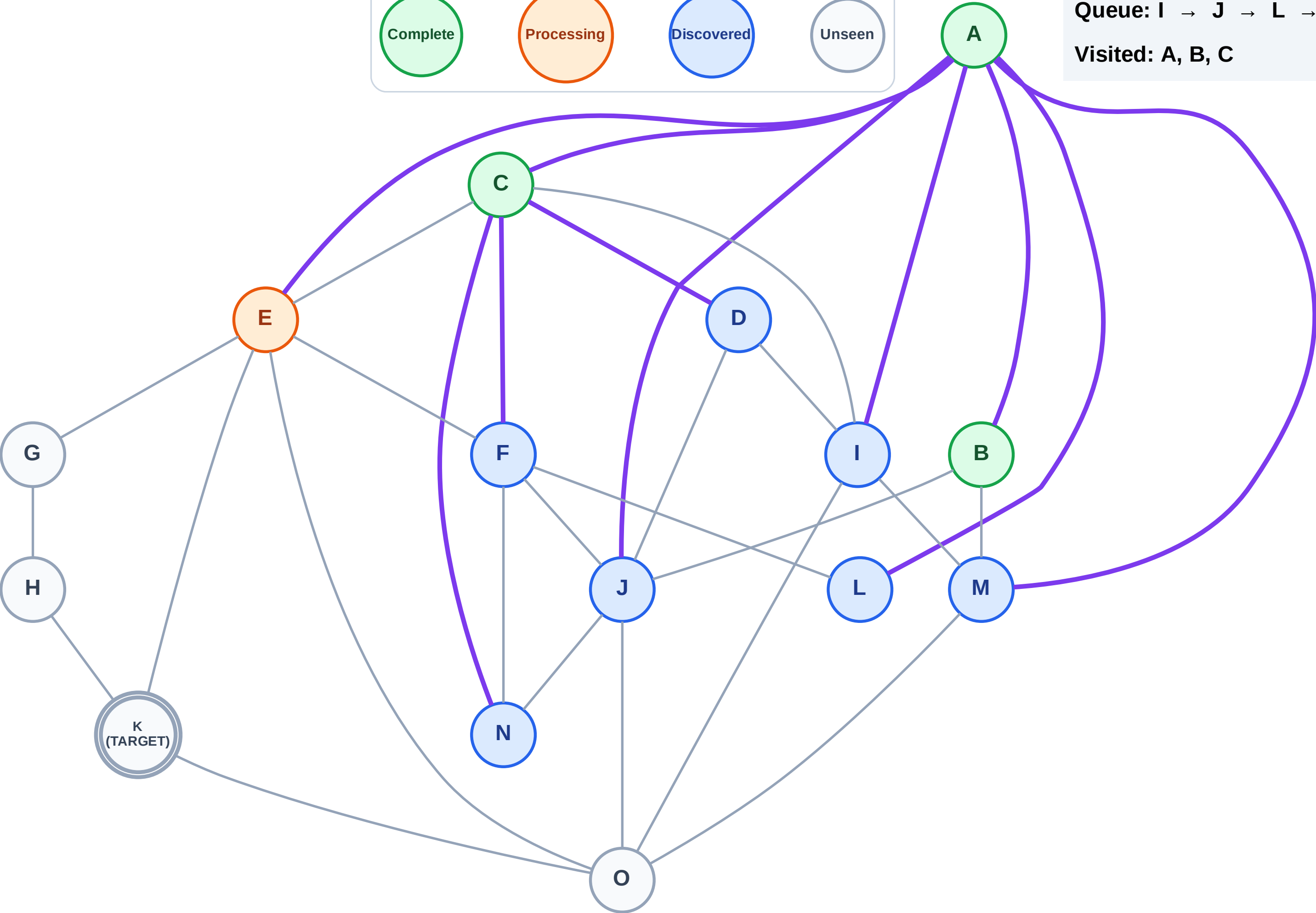
Processing

Discovered

Unseen

Queue: I → J → L → M → D → F → N

Visited: A, B, C



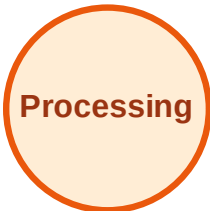
BFS Traversal

Step 9: Discover G, K, O from E

Legend



Complete



Processing



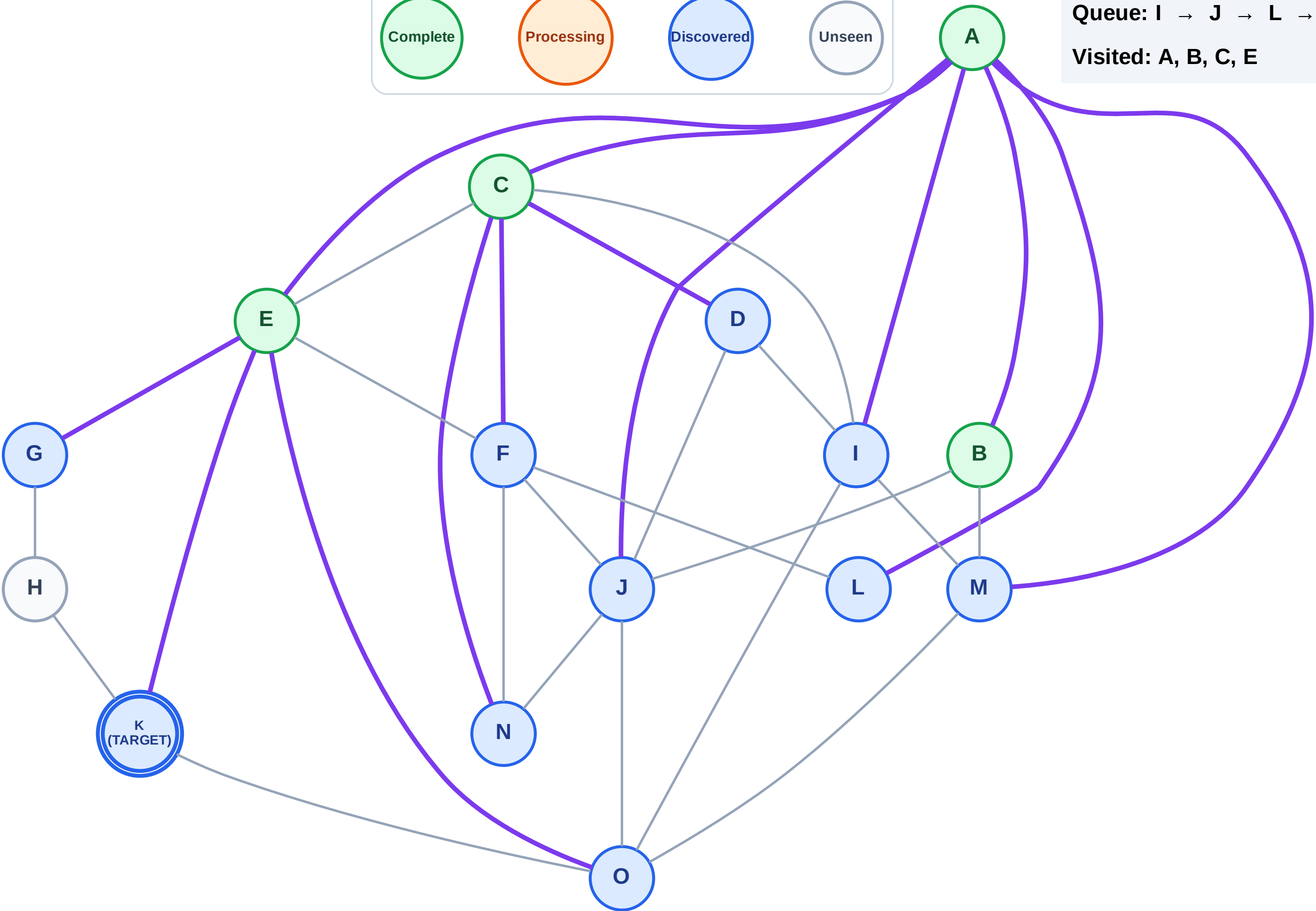
Discovered



Unseen

Queue: I → J → L → M → D → F → N → G → K → O

Visited: A, B, C, E



BFS Traversal

Step 10: Process node I

Legend

Complete

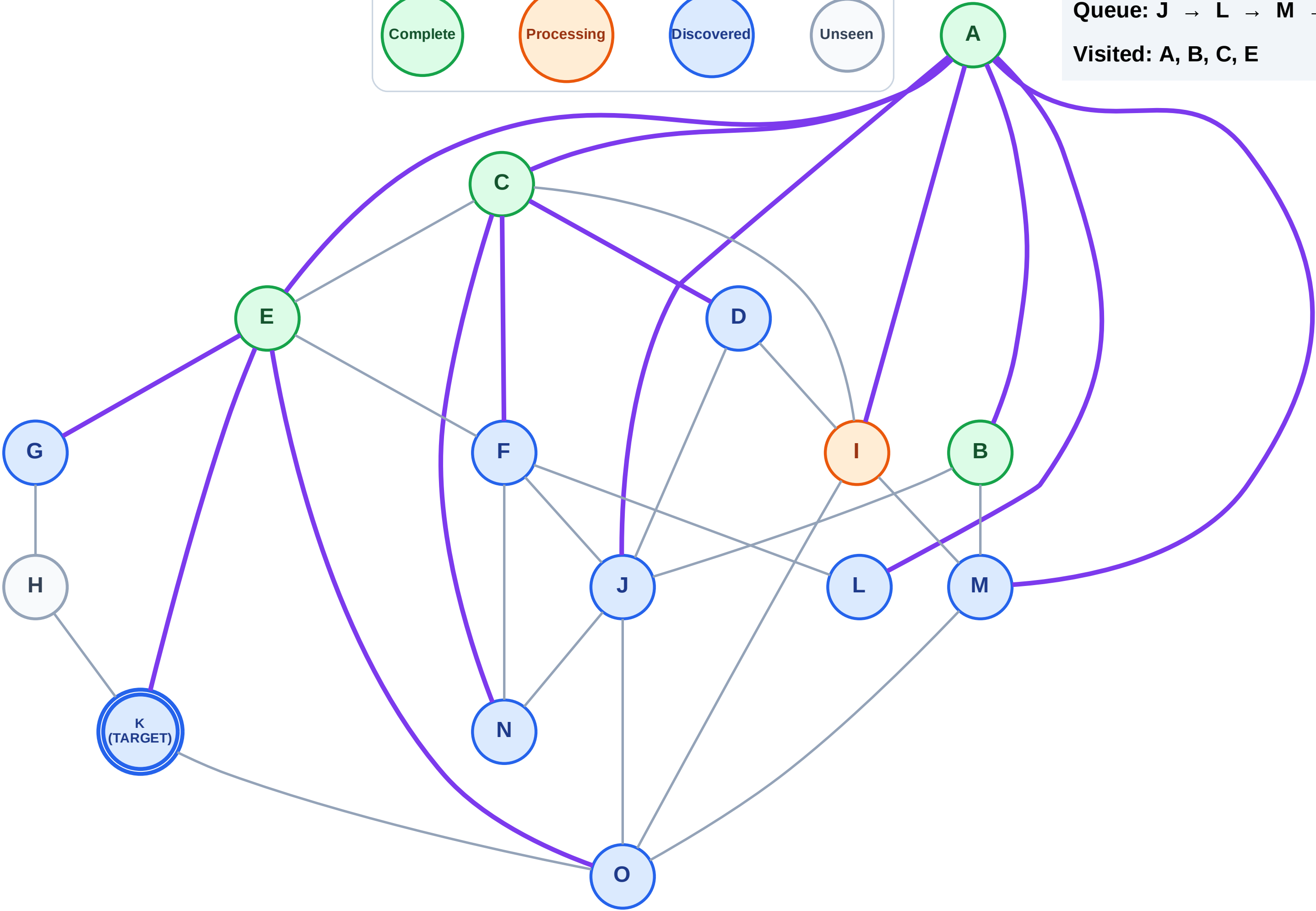
Processing

Discovered

Unseen

Queue: J → L → M → D → F → N → G → K → O

Visited: A, B, C, E



BFS Traversal

Step 11: No new neighbors from I

Legend



Complete



Processing



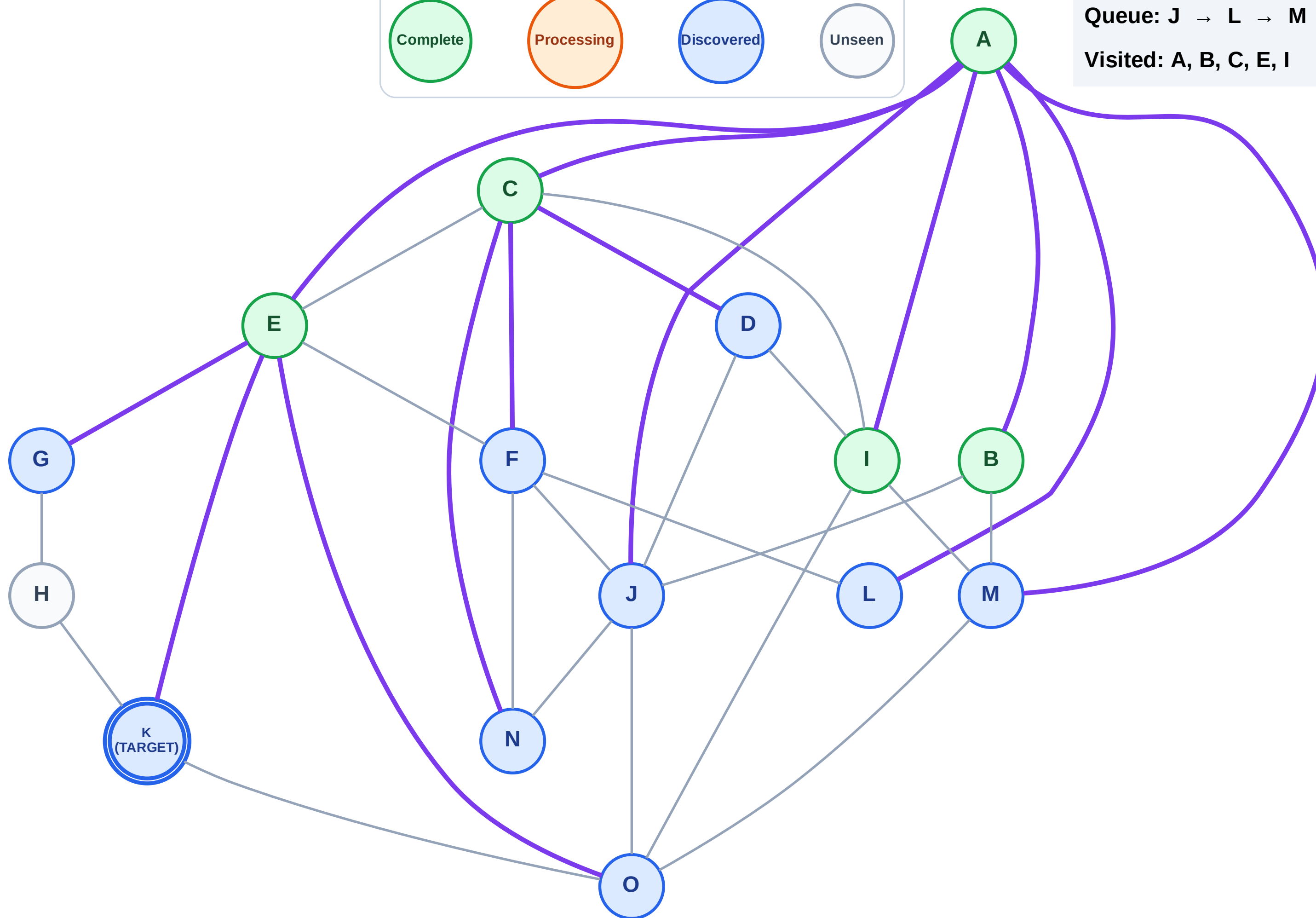
Discovered



Unseen

Queue: J → L → M → D → F → N → G → K → O

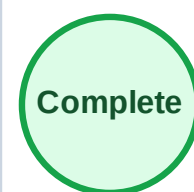
Visited: A, B, C, E, I



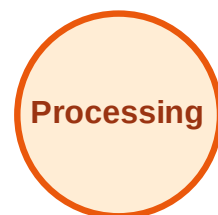
BFS Traversal

Step 12: Process node J

Legend



Complete



Processing



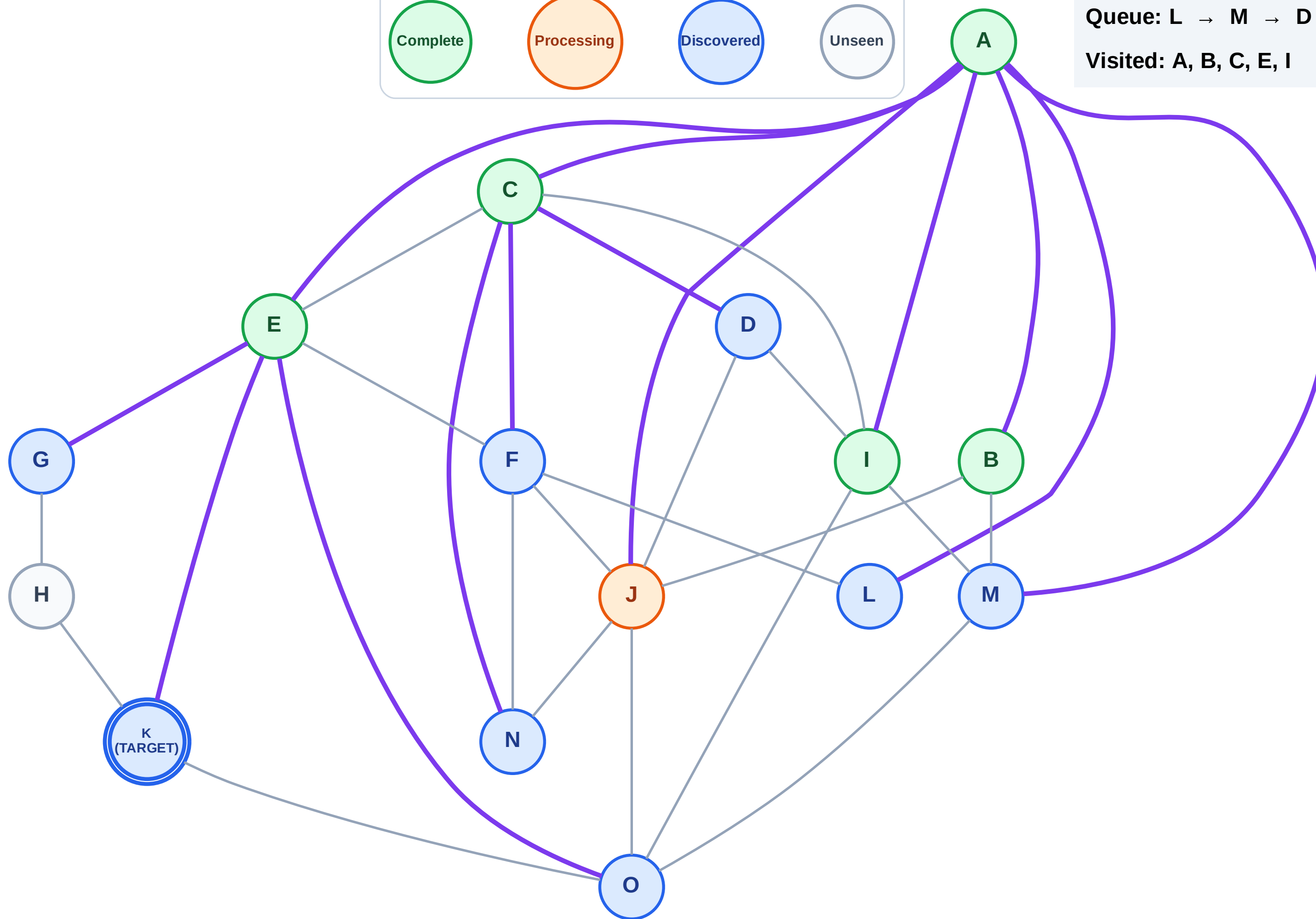
Discovered



Unseen

Queue: L → M → D → F → N → G → K → O

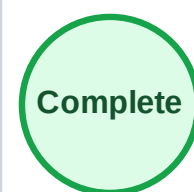
Visited: A, B, C, E, I



BFS Traversal

Step 13: No new neighbors from J

Legend



Complete



Processing



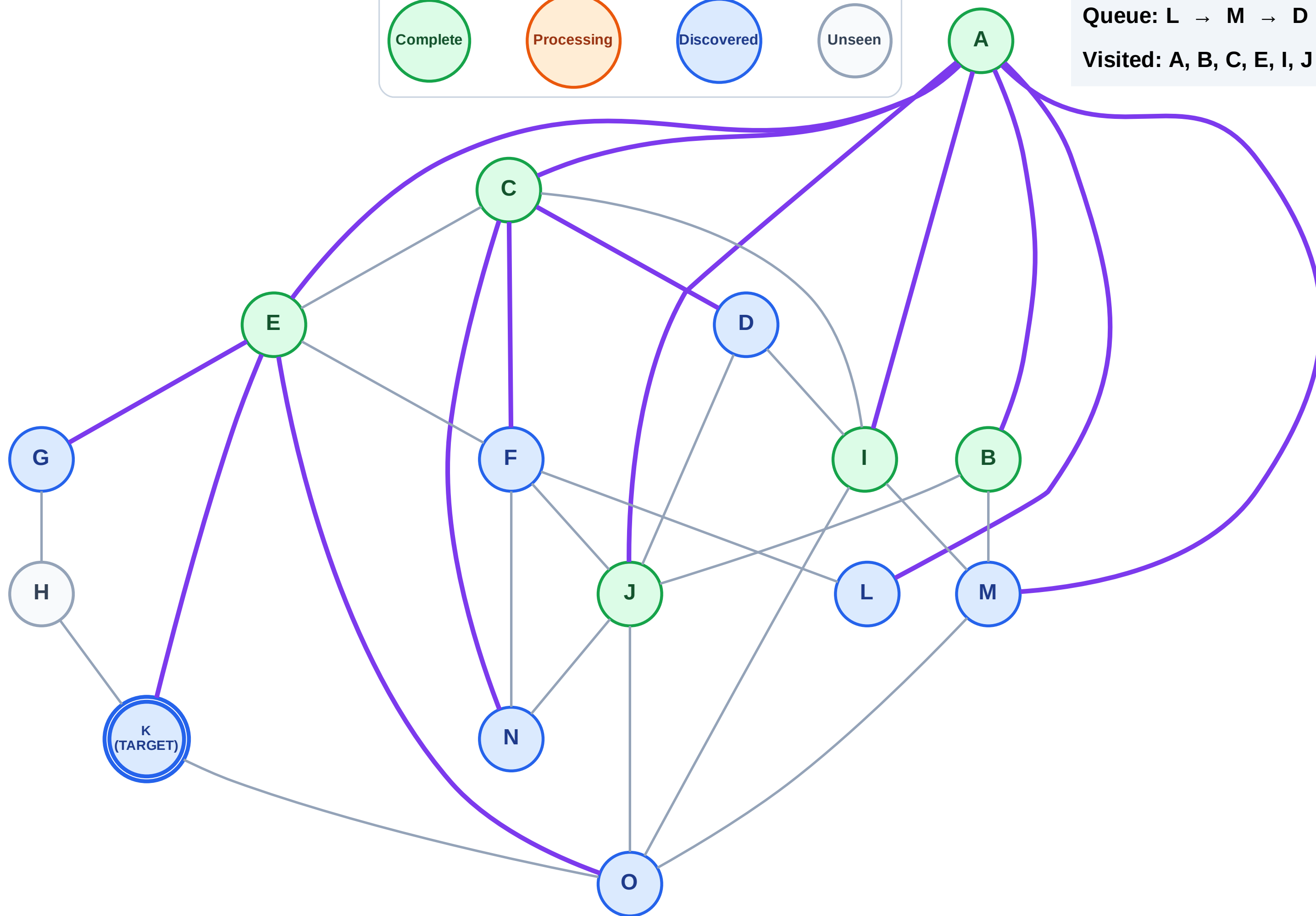
Discovered



Unseen

Queue: L → M → D → F → N → G → K → O

Visited: A, B, C, E, I, J



BFS Traversal

Step 14: Process node L

Legend

Complete

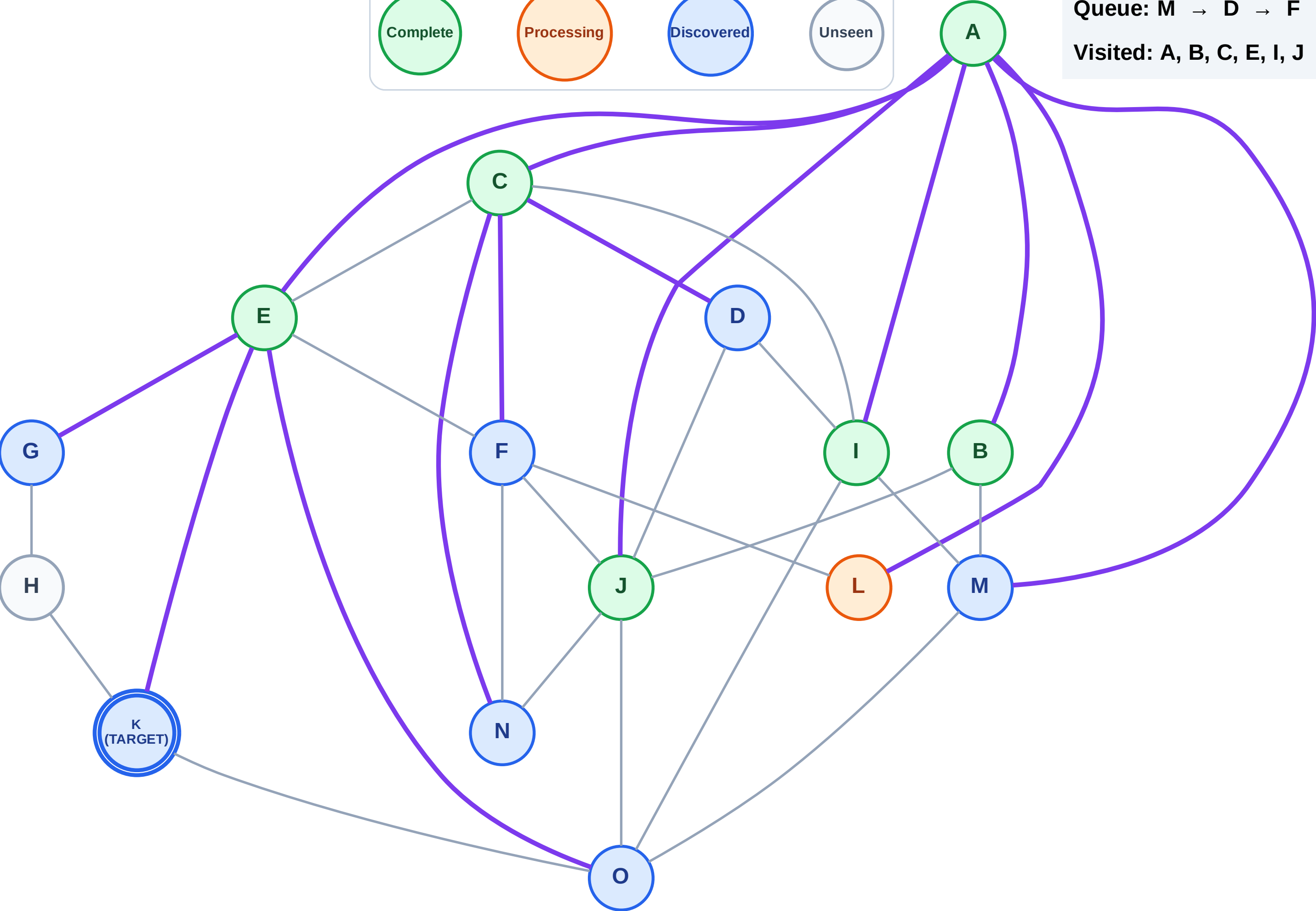
Processing

Discovered

Unseen

Queue: M → D → F → N → G → K → O

Visited: A, B, C, E, I, J



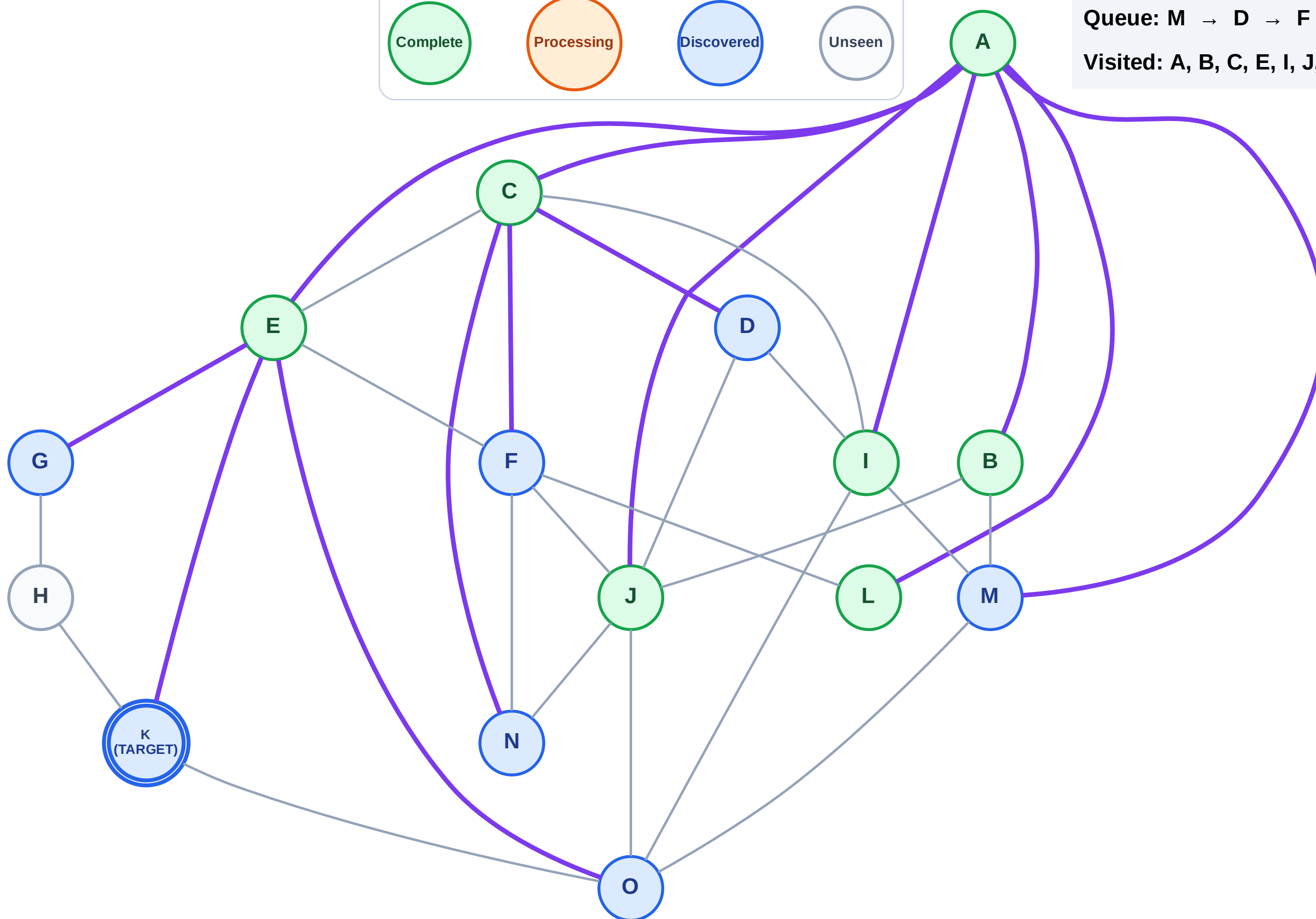
Step 15: No new neighbors from L

Legend



Complete Processing Discovered Unseen

Visited: A, B, C, E, I, J, L



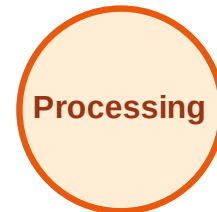
BFS Traversal

Step 16: Process node M

Legend



Complete



Processing

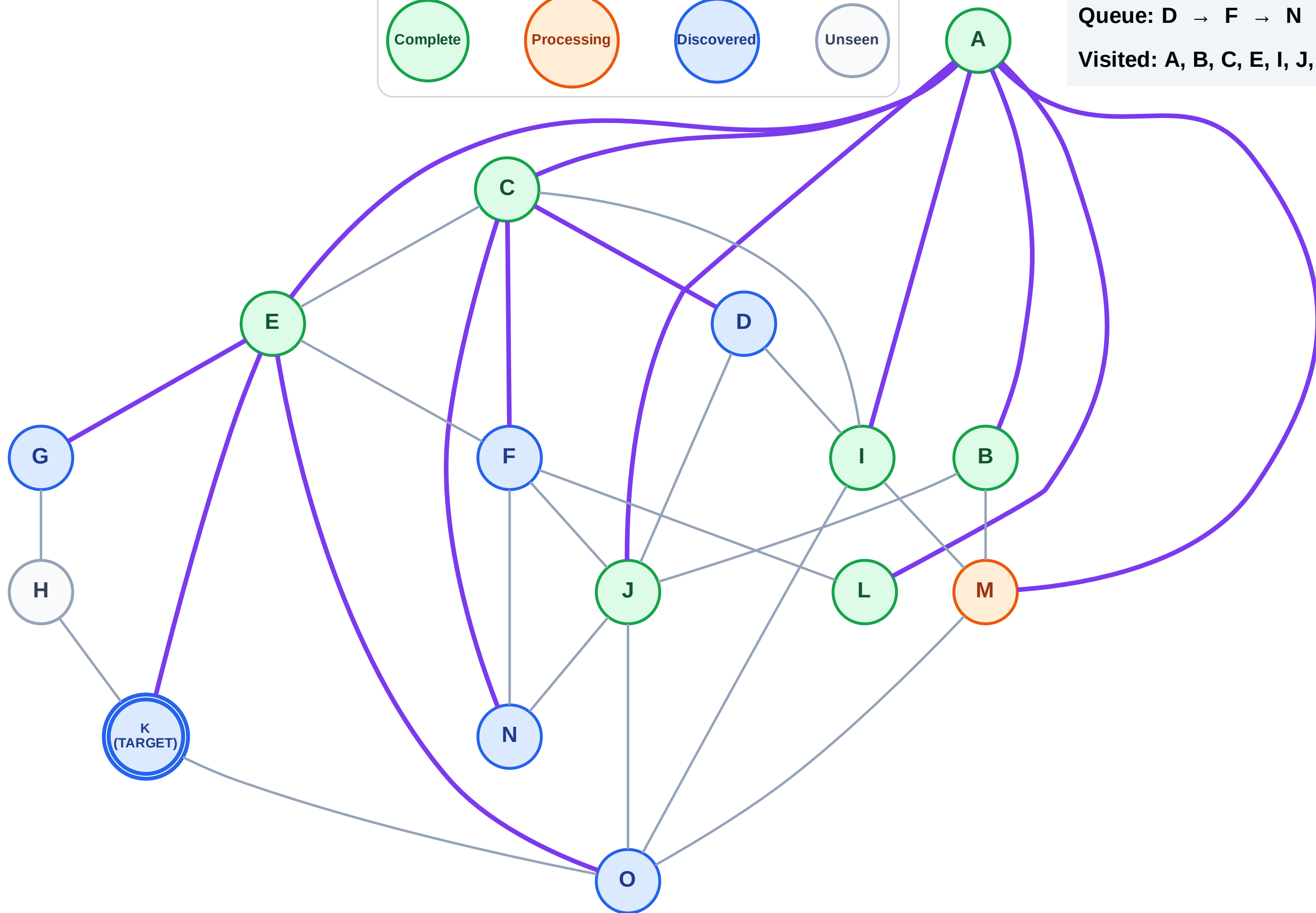


Discovered



Unseen

Queue: D → F → N → G → K → O
Visited: A, B, C, E, I, J, L



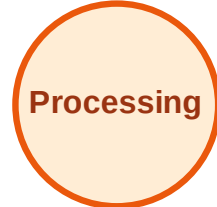
BFS Traversal

Step 17: No new neighbors from M

Legend



Complete



Processing

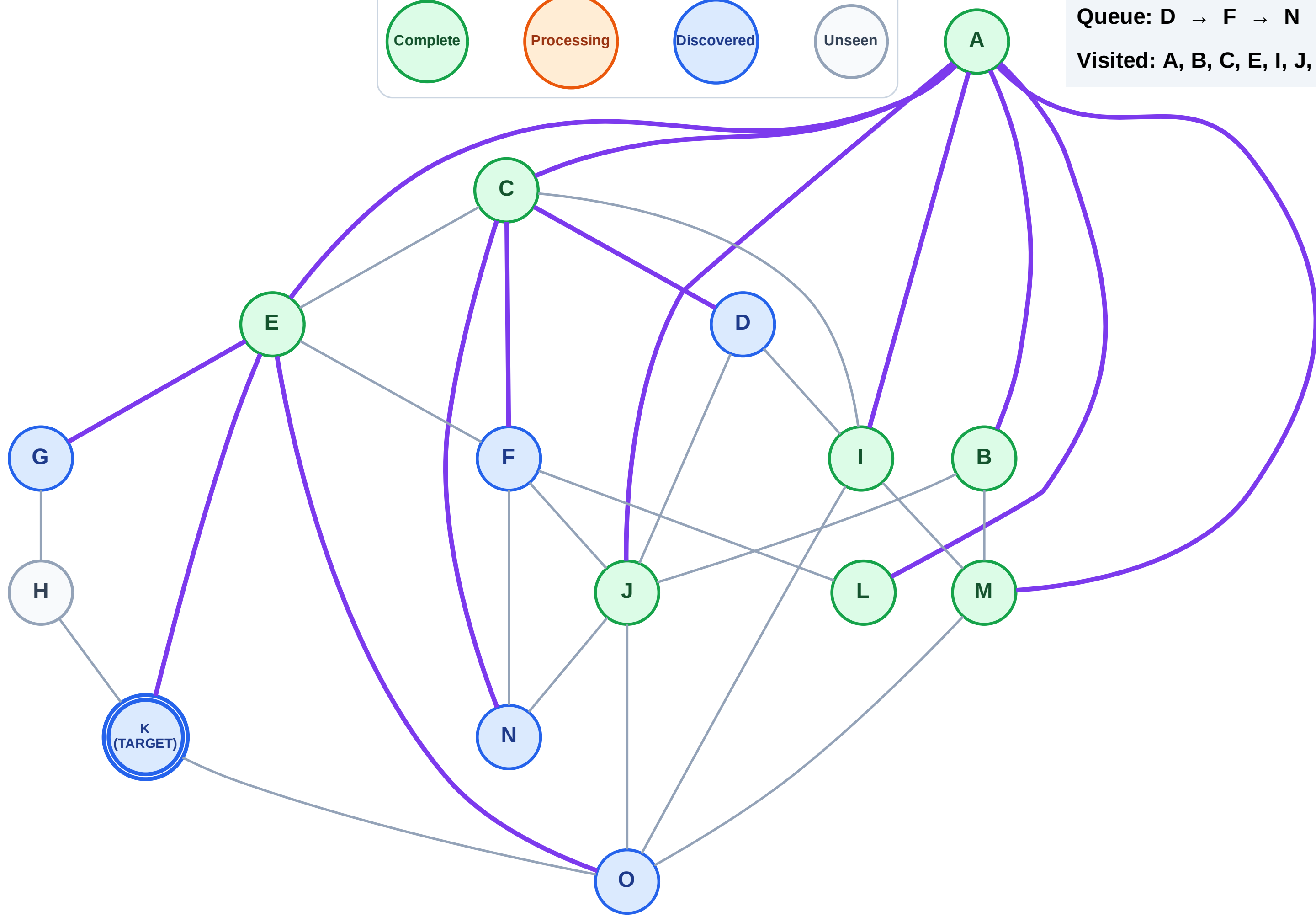


Discovered



Unseen

Queue: D → F → N → G → K → O
Visited: A, B, C, E, I, J, L, M



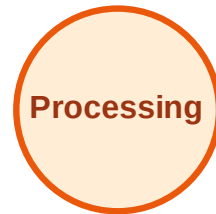
BFS Traversal

Step 18: Process node D

Legend



Complete



Processing



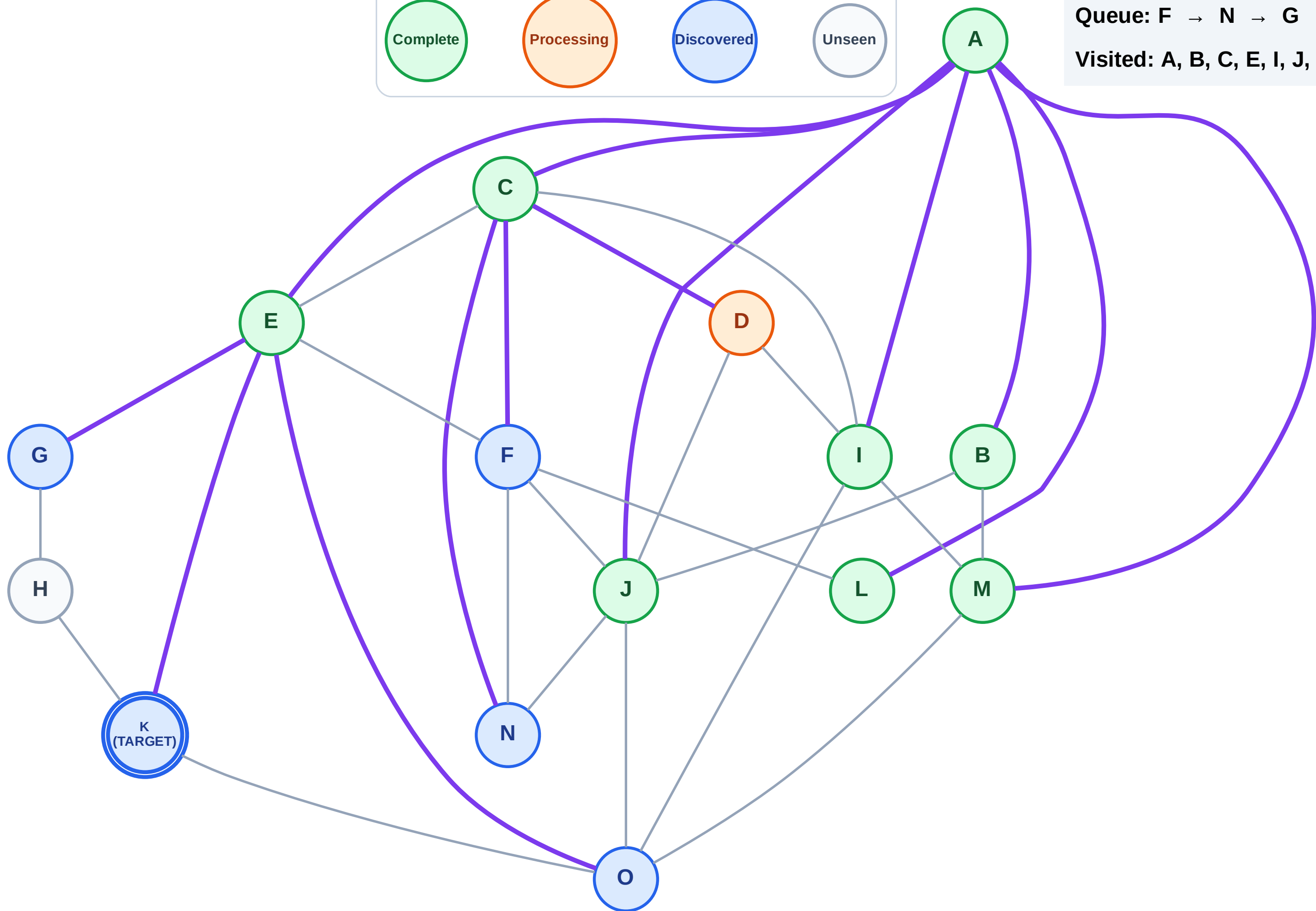
Discovered



Unseen

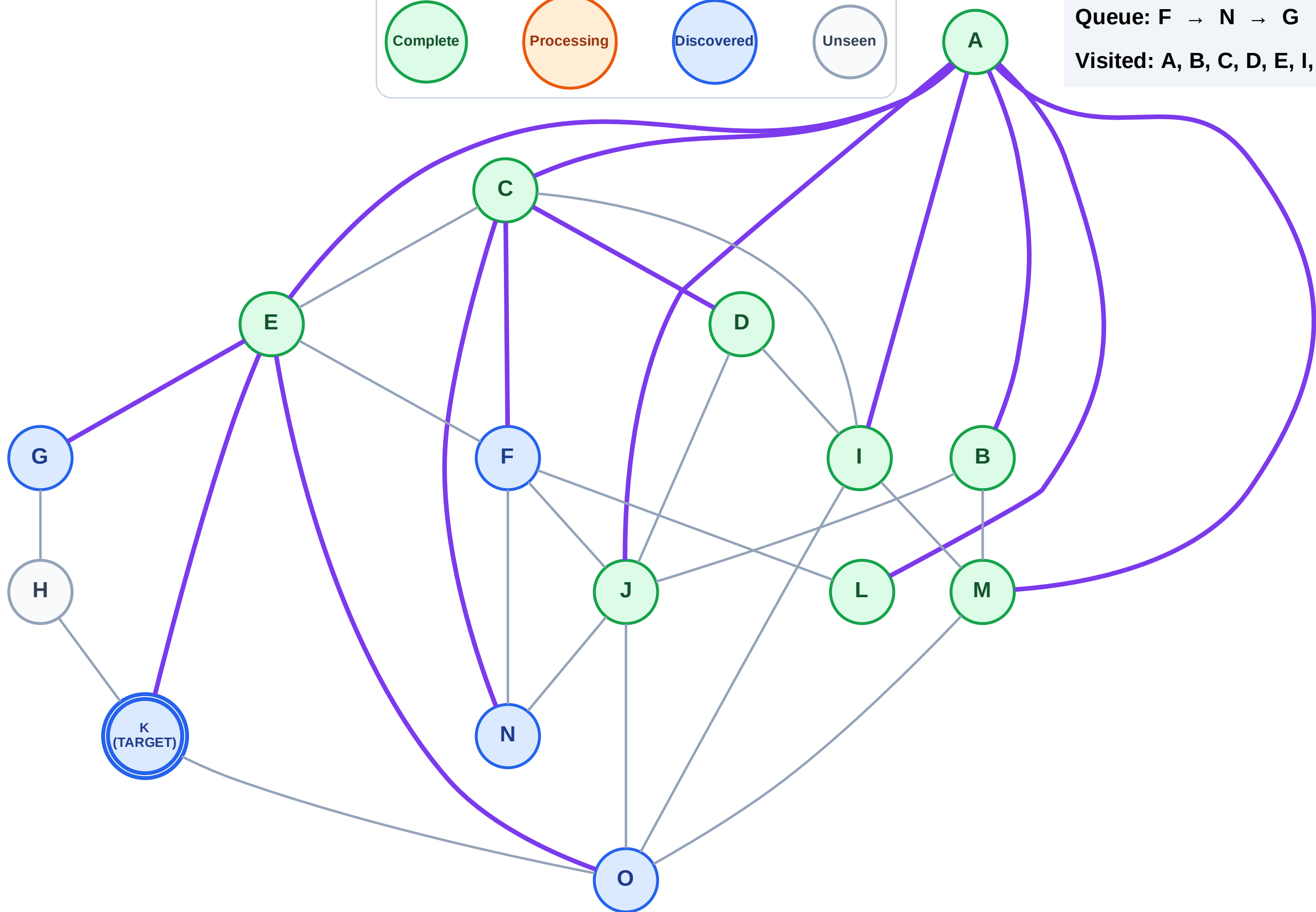
Queue: F → N → G → K → O

Visited: A, B, C, E, I, J, L, M



Step 19: No new neighbors from D

Step 19: No new neighbors from D



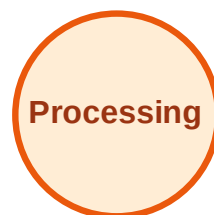
BFS Traversal

Step 20: Process node F

Legend



Complete



Processing



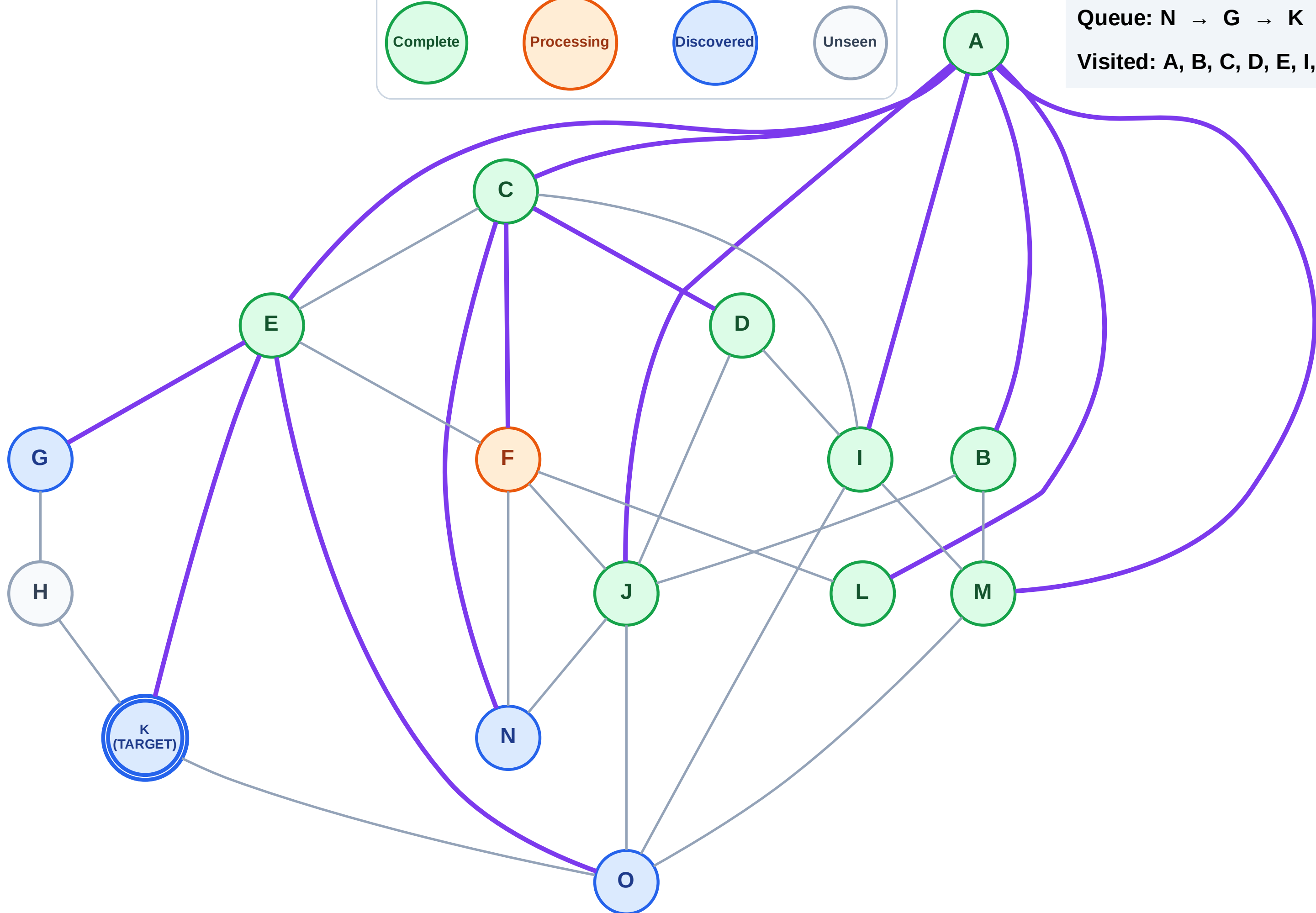
Discovered



Unseen

Queue: N → G → K → O

Visited: A, B, C, D, E, I, J, L, M



BFS Traversal

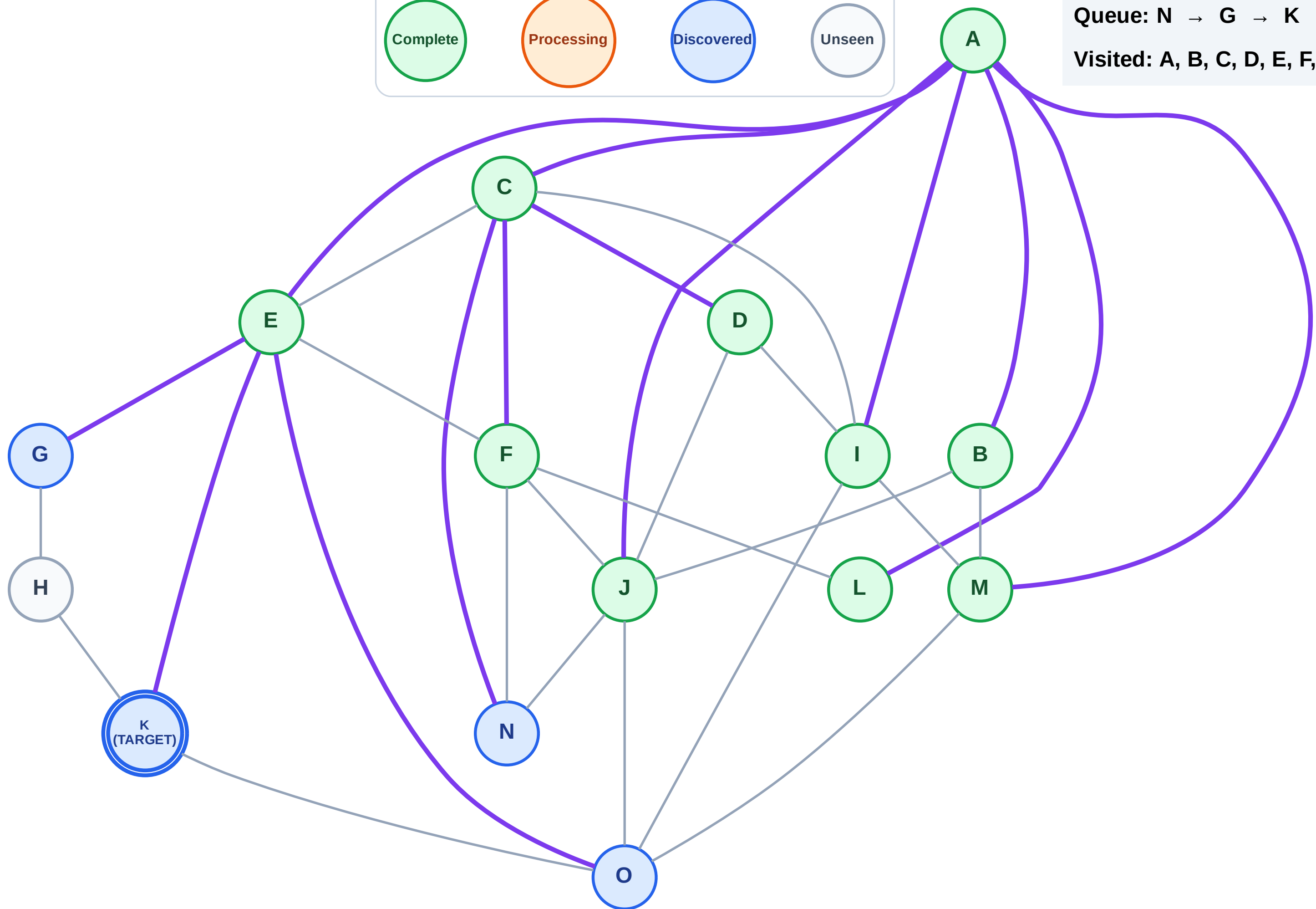
Step 21: No new neighbors from F

Legend



Queue: N → G → K → O

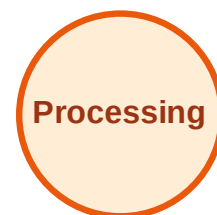
Visited: A, B, C, D, E, F, I, J, L, M



BFS Traversal

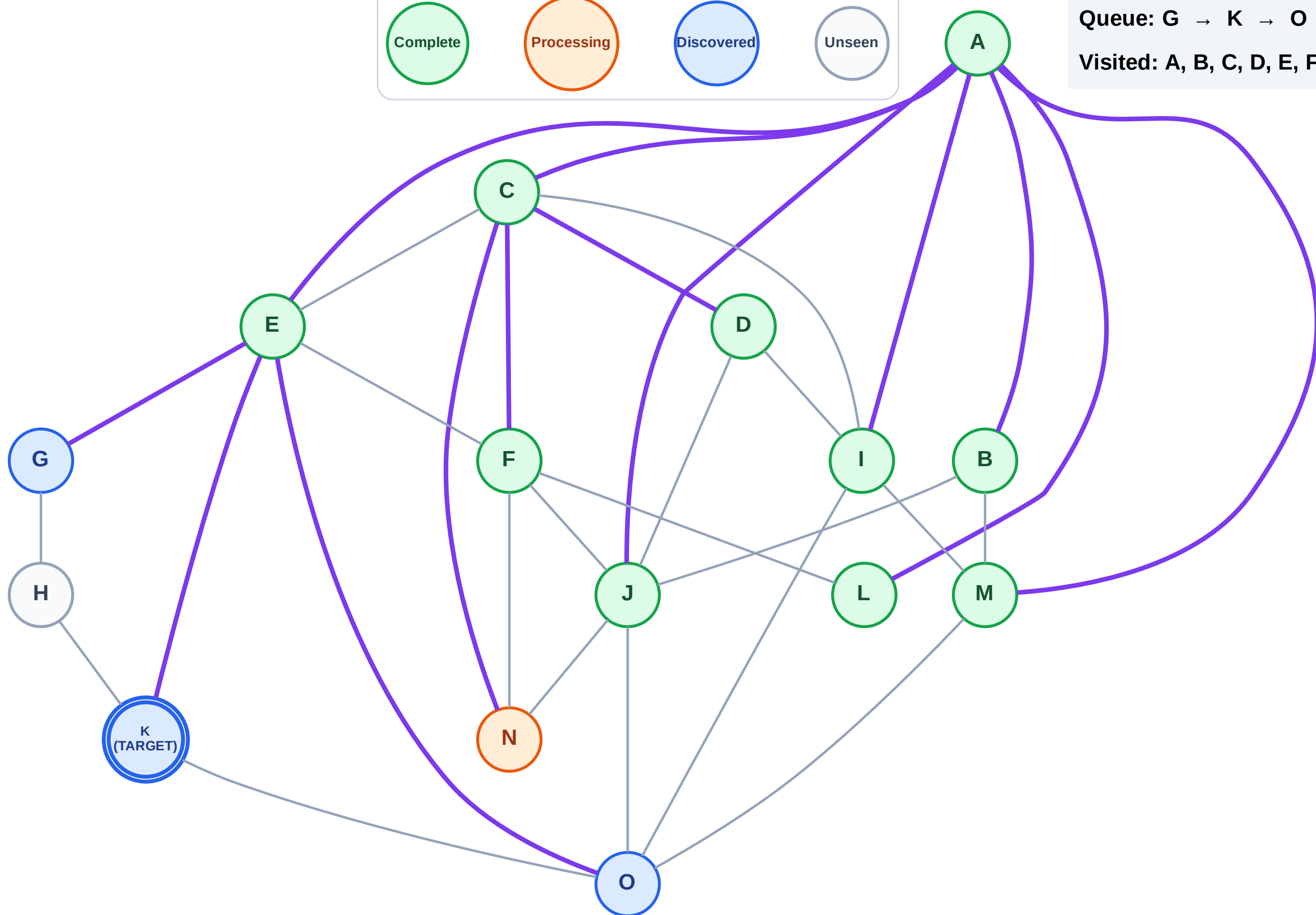
Step 22: Process node N

Legend



Queue: G → K → O

Visited: A, B, C, D, E, F, I, J, L, M



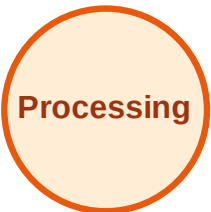
BFS Traversal

Step 23: No new neighbors from N

Legend



Complete



Processing



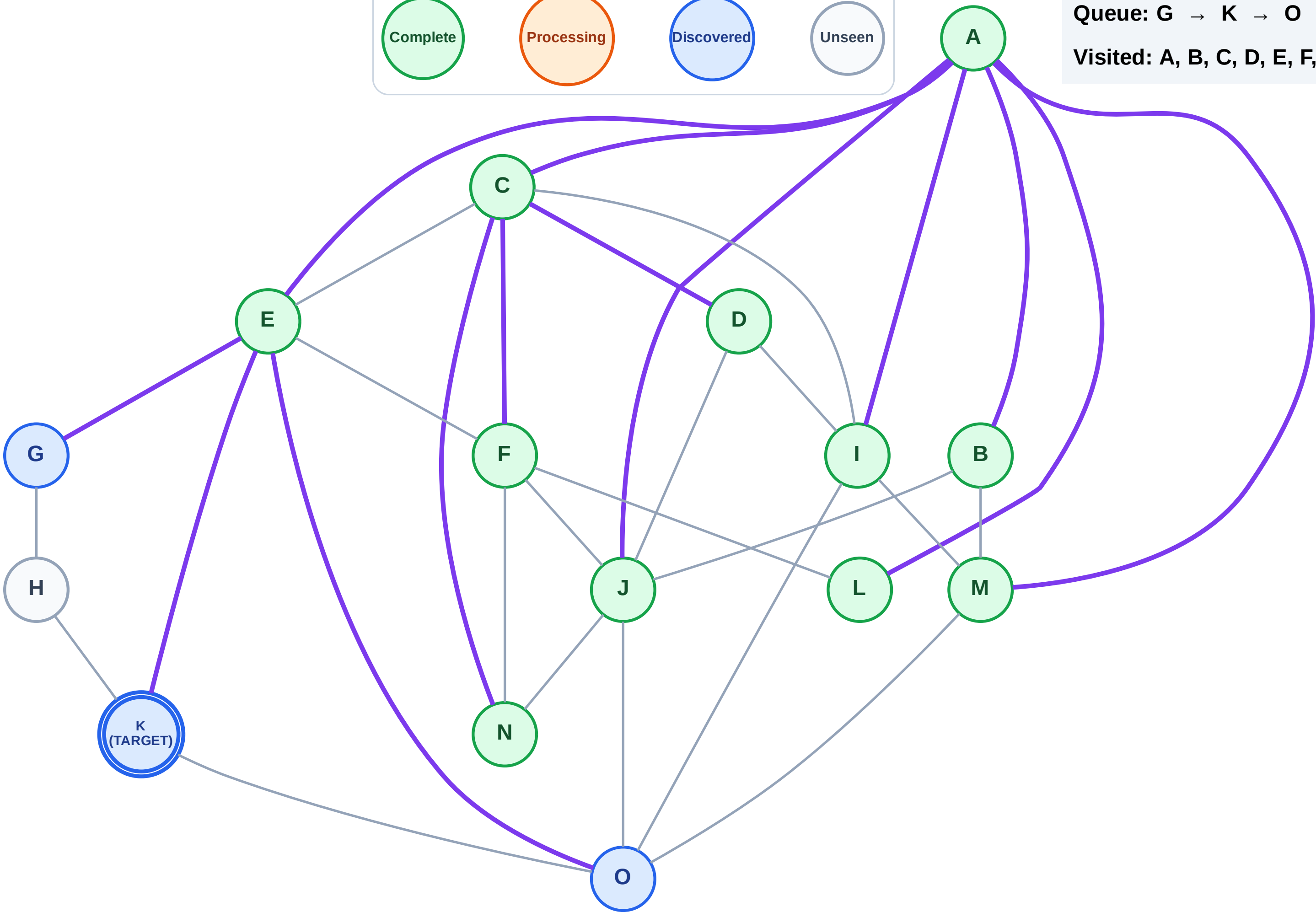
Discovered



Unseen

Queue: G → K → O

Visited: A, B, C, D, E, F, I, J, L, M, N



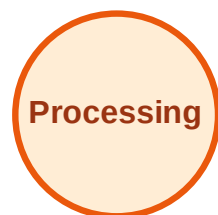
BFS Traversal

Step 24: Process node G

Legend



Complete



Processing



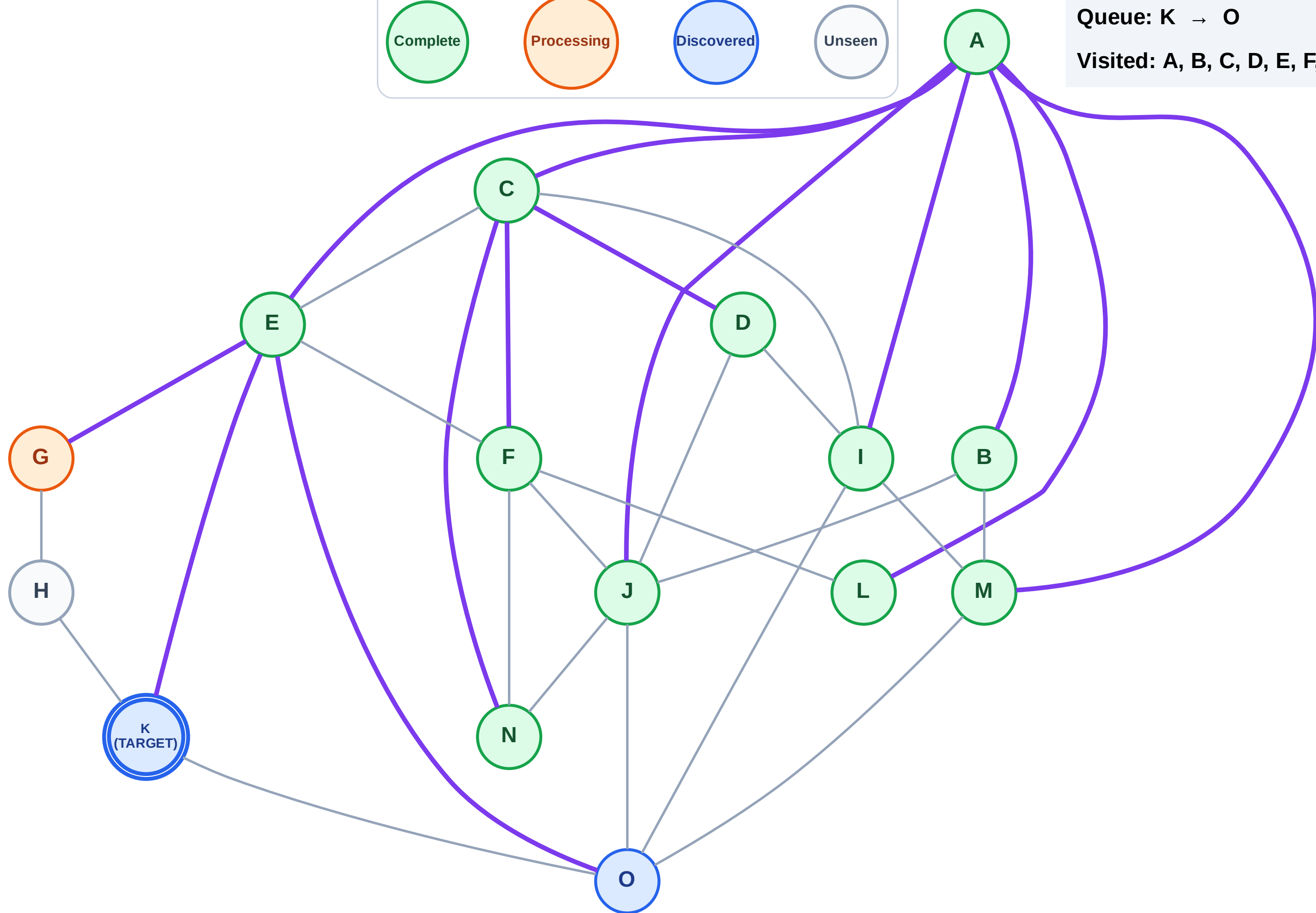
Discovered



Unseen

Queue: K → O

Visited: A, B, C, D, E, F, I, J, L, M, N



Step 25: Discover H from G

Step 25: Discover H from G

Legend

Complete

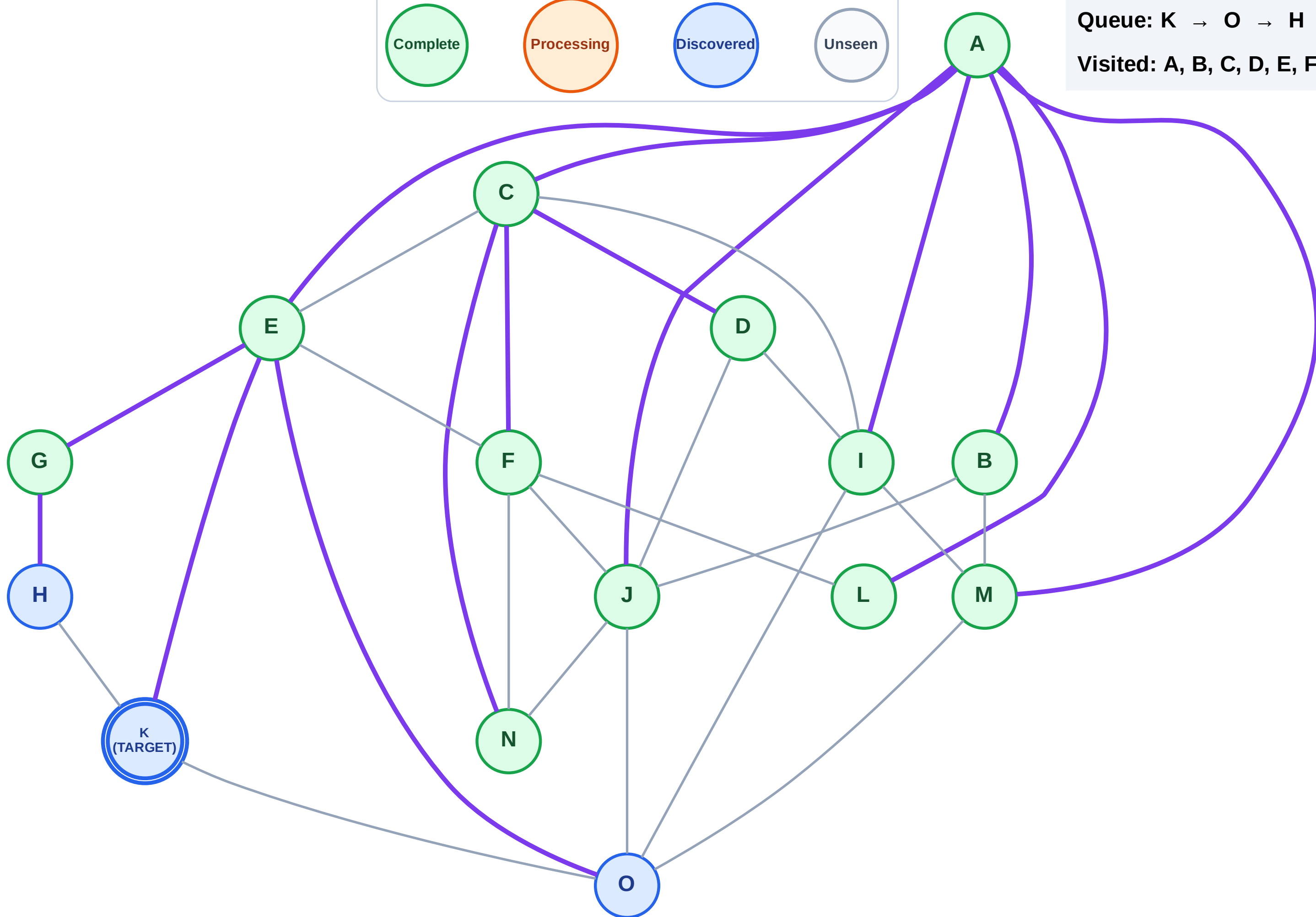
Processing

Discovered

Unseen

Queue: K → O → H

Visited: A, B, C, D, E, F, G, I, J, L, M, N



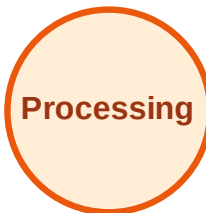
BFS Traversal

Step 26: Process node K

Legend



Complete



Processing



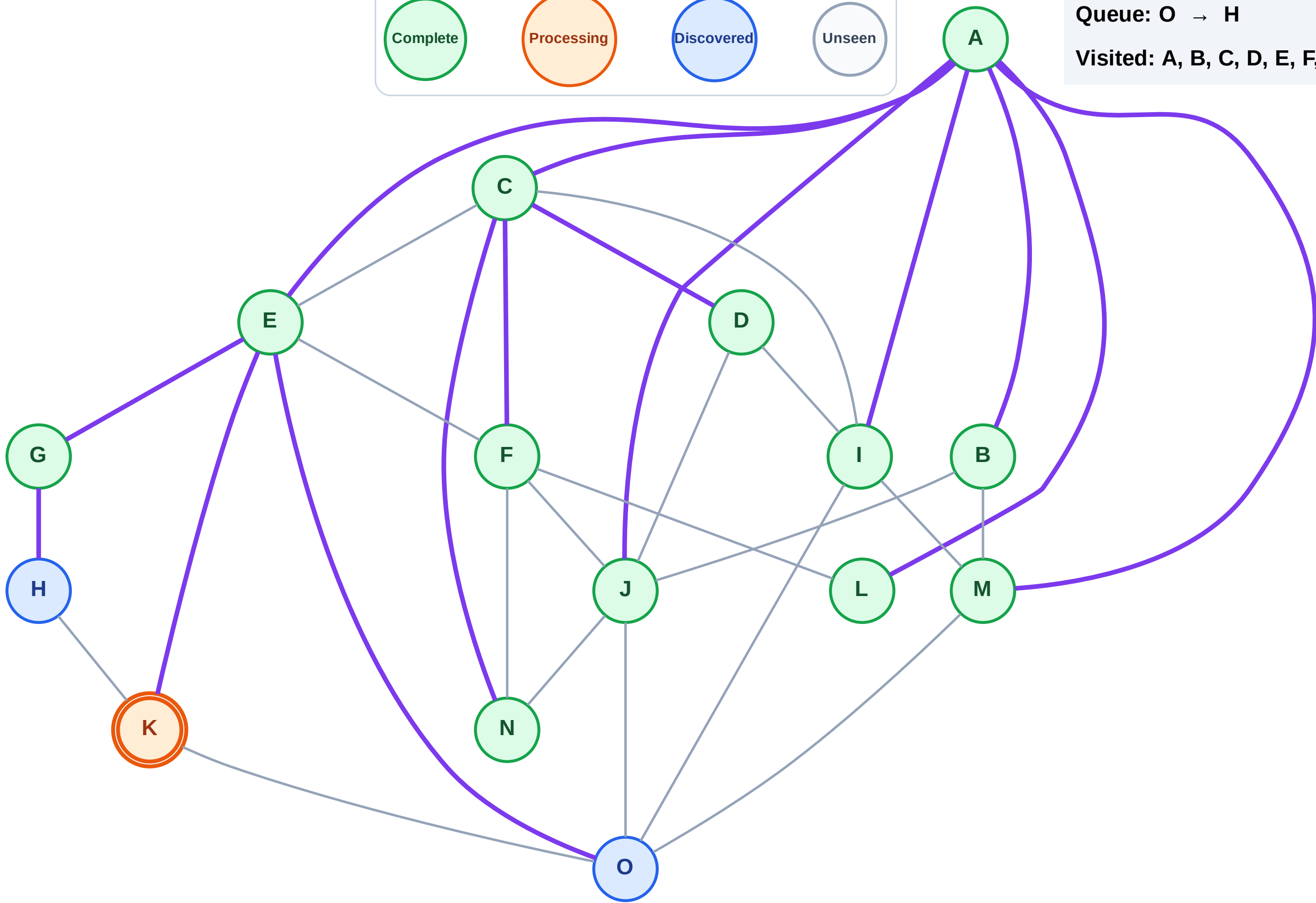
Discovered



Unseen

Queue: O → H

Visited: A, B, C, D, E, F, G, I, J, L, M, N



BFS Traversal

Step 27: TARGET NODE K REACHED! Path (2 hops): A → E → K

Legend

Complete

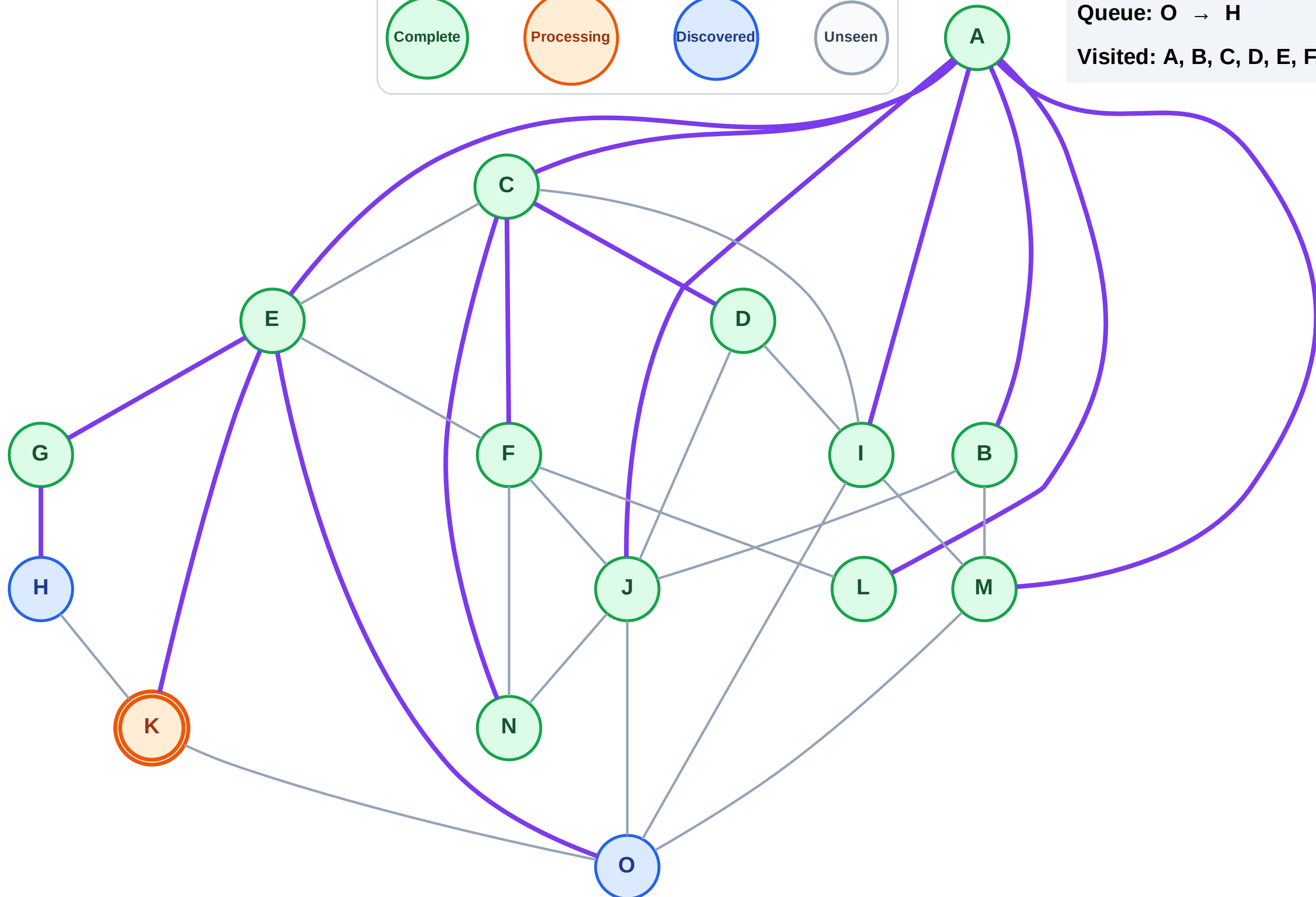
Processing

Discovered

Unseen

Queue: O → H

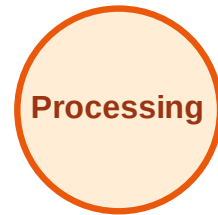
Visited: A, B, C, D, E, F, G, I, J, K, L, M, N



DFS Traversal

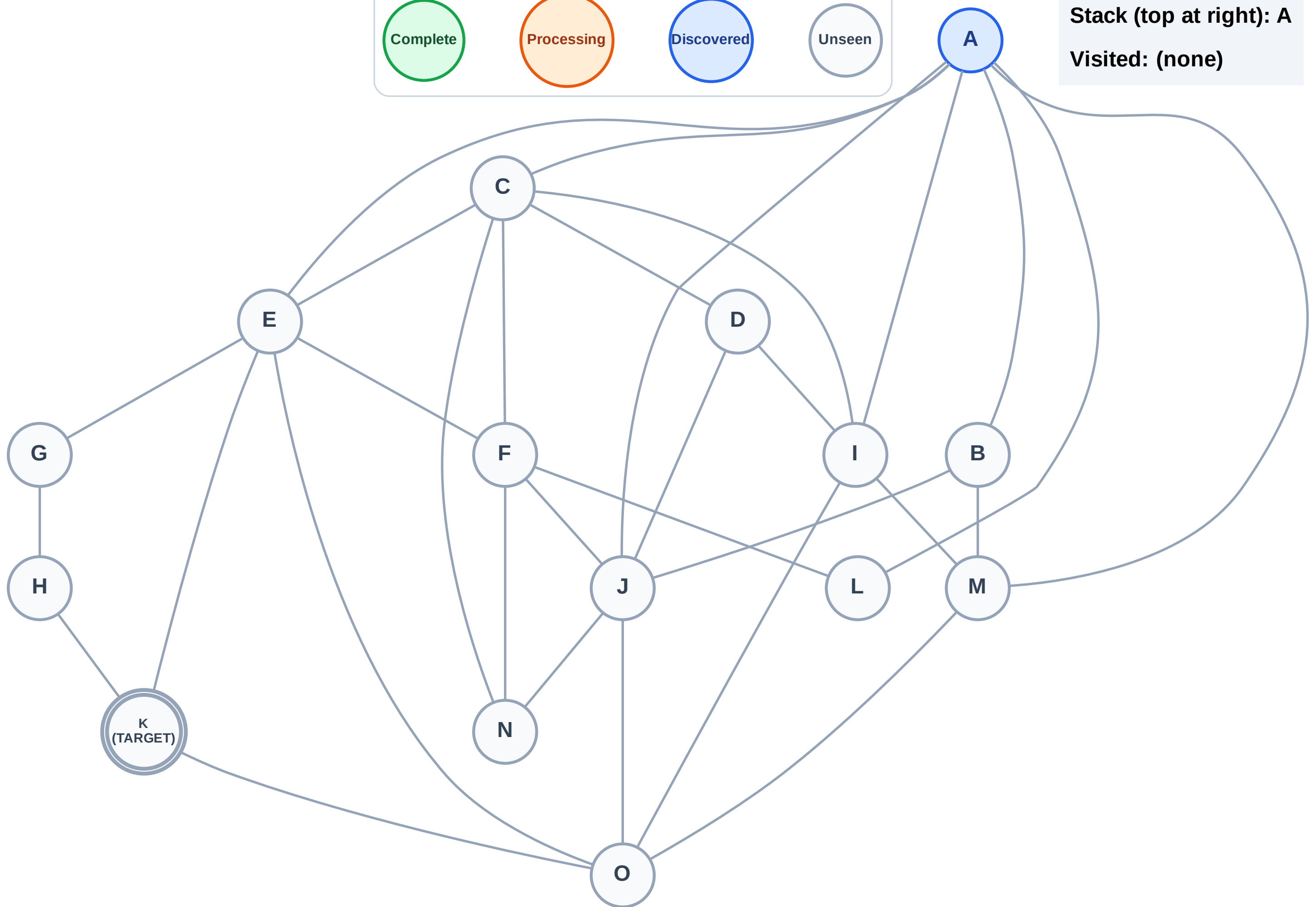
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



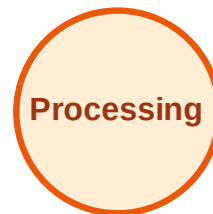
DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



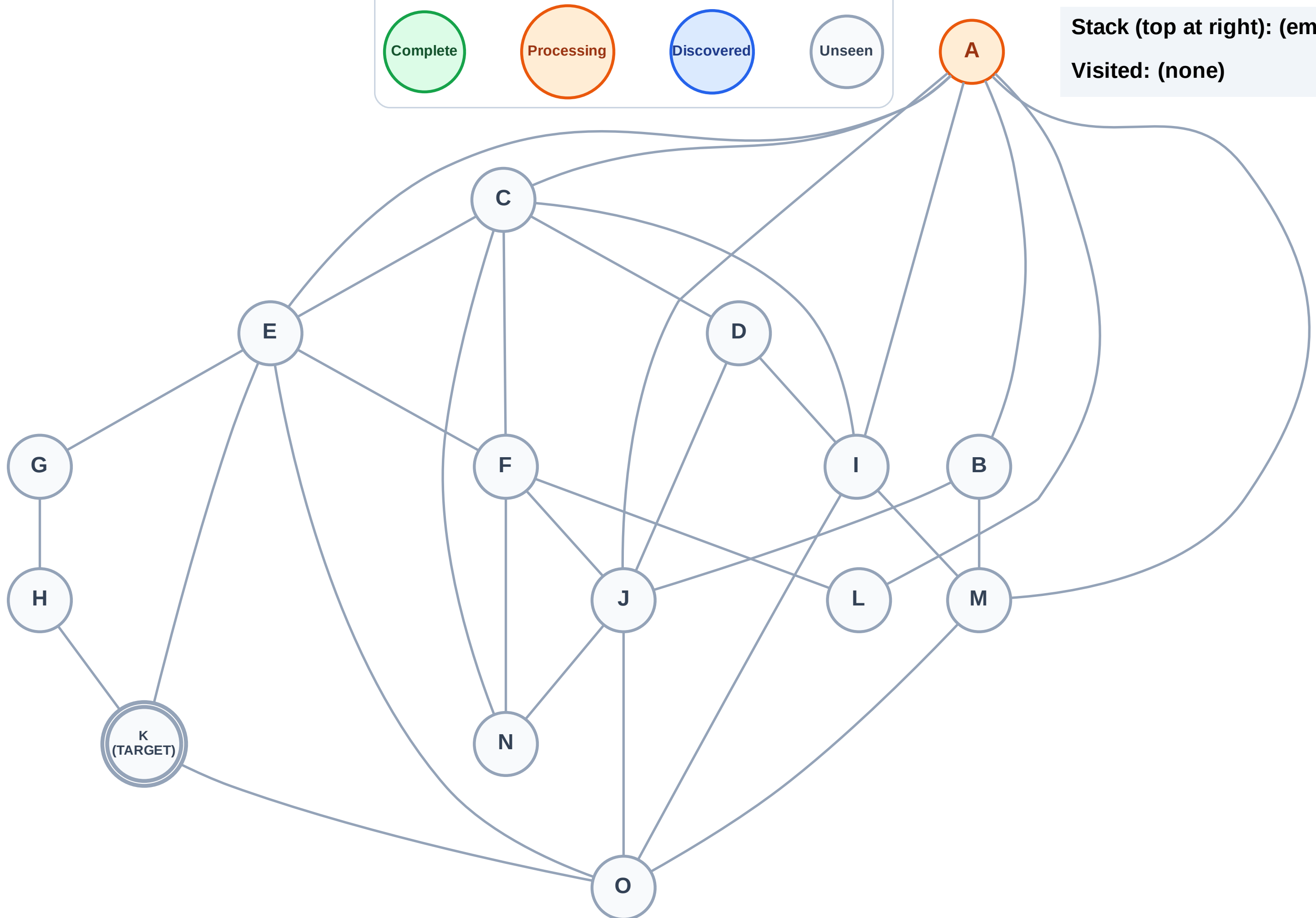
Discovered



Unseen

Stack (top at right): (empty)

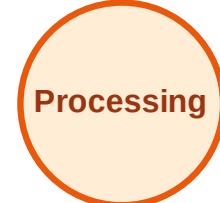
Visited: (none)



DFS Traversal

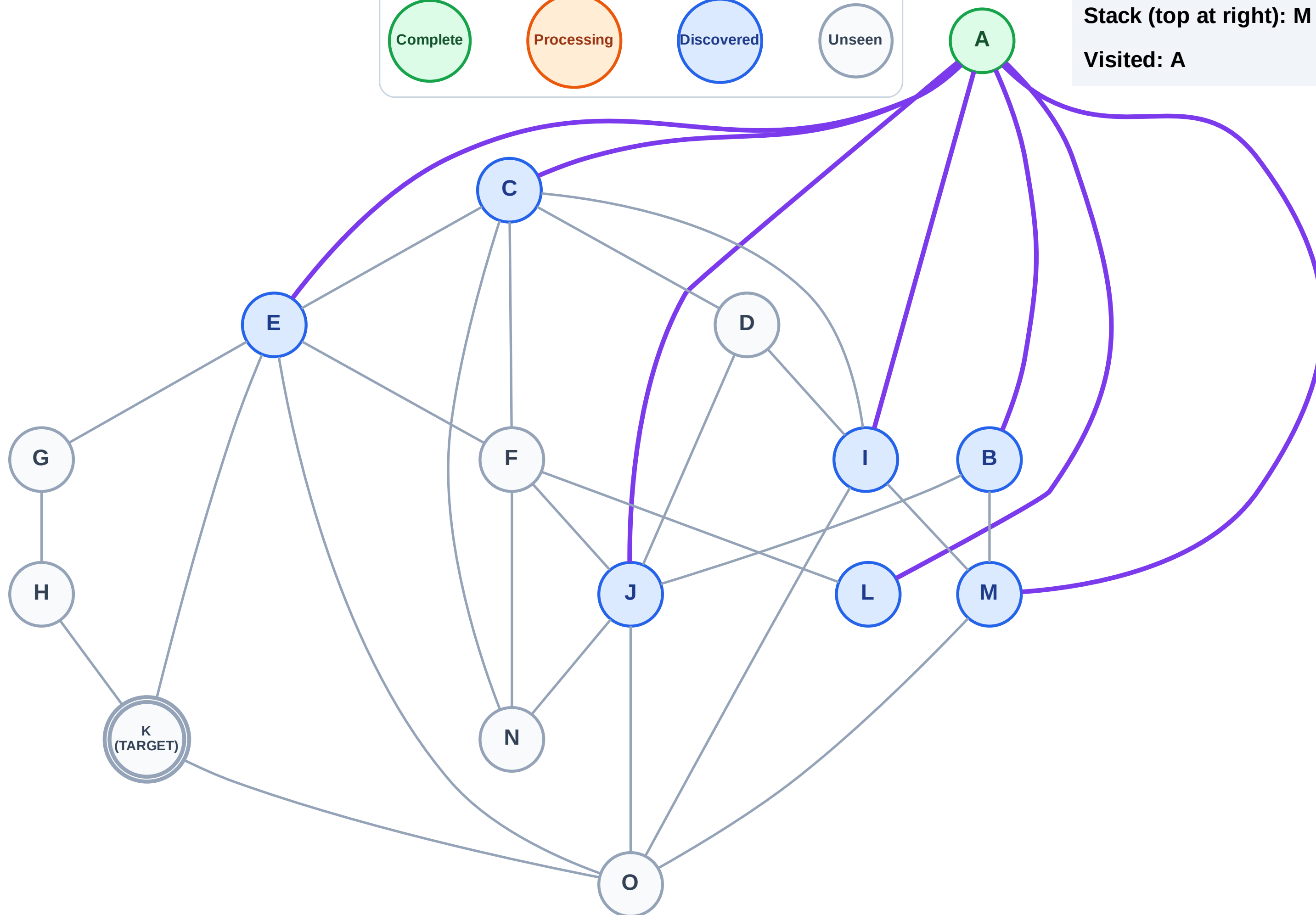
Step 3: Discover M, L, J, I, E, C, B from A

Legend



Stack (top at right): M | L | J | I | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

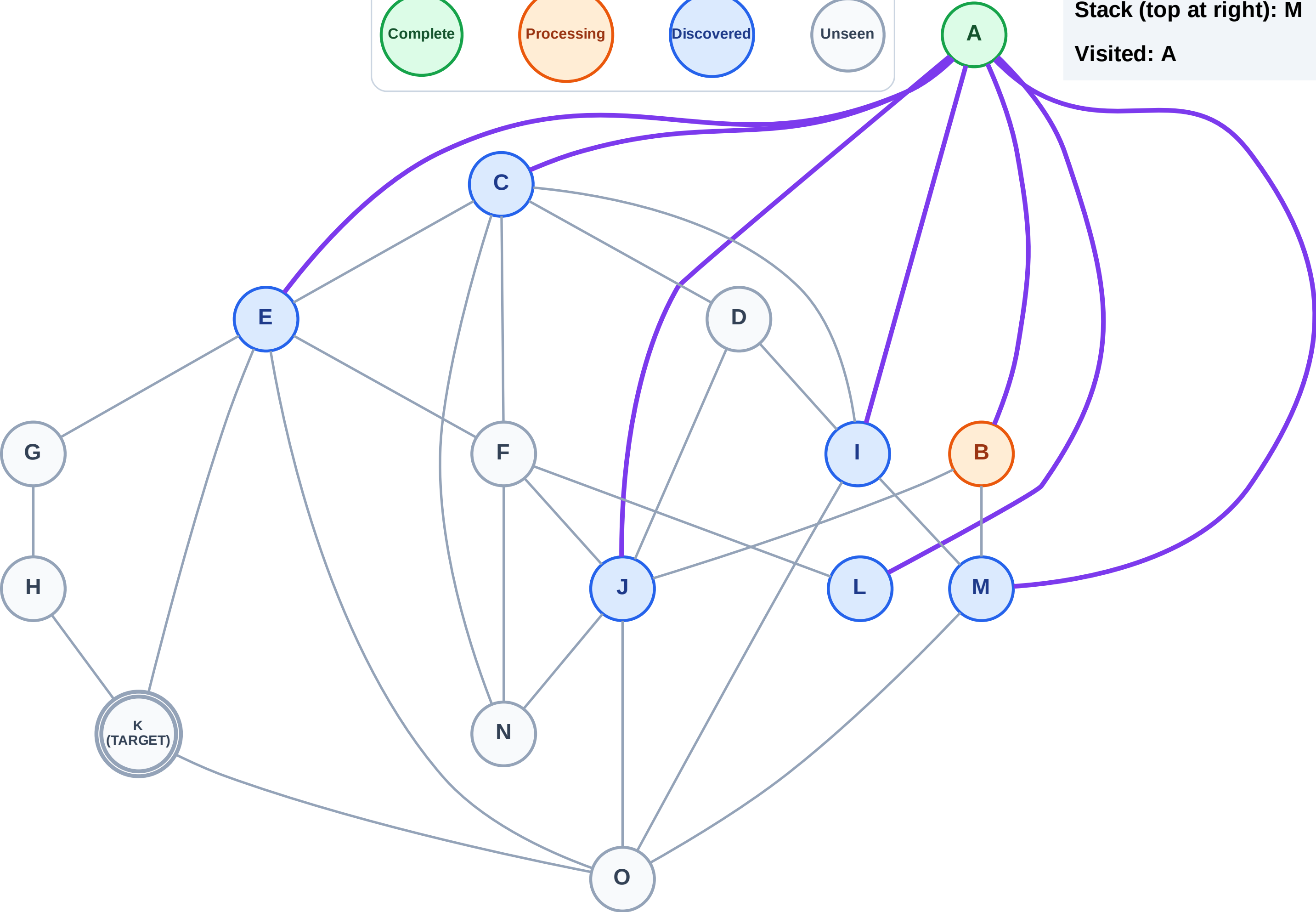
Processing

Discovered

Unseen

Stack (top at right): M | L | J | I | E | C

Visited: A



DFS Traversal

Step 5: No new neighbors from B

Legend

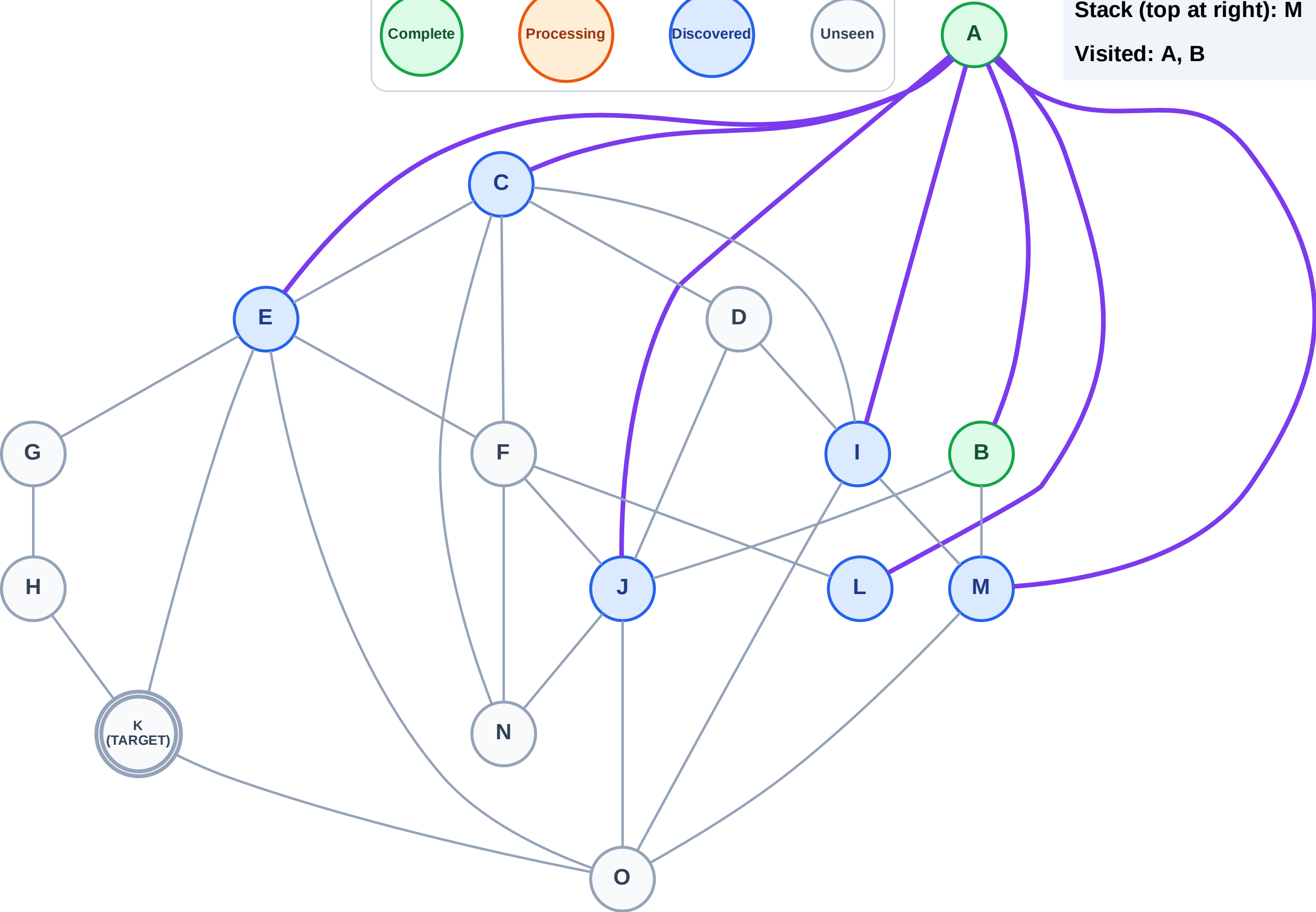
Complete

Processing

Discovered

Unseen

Stack (top at right): M | L | J | I | E | C
Visited: A, B



DFS Traversal

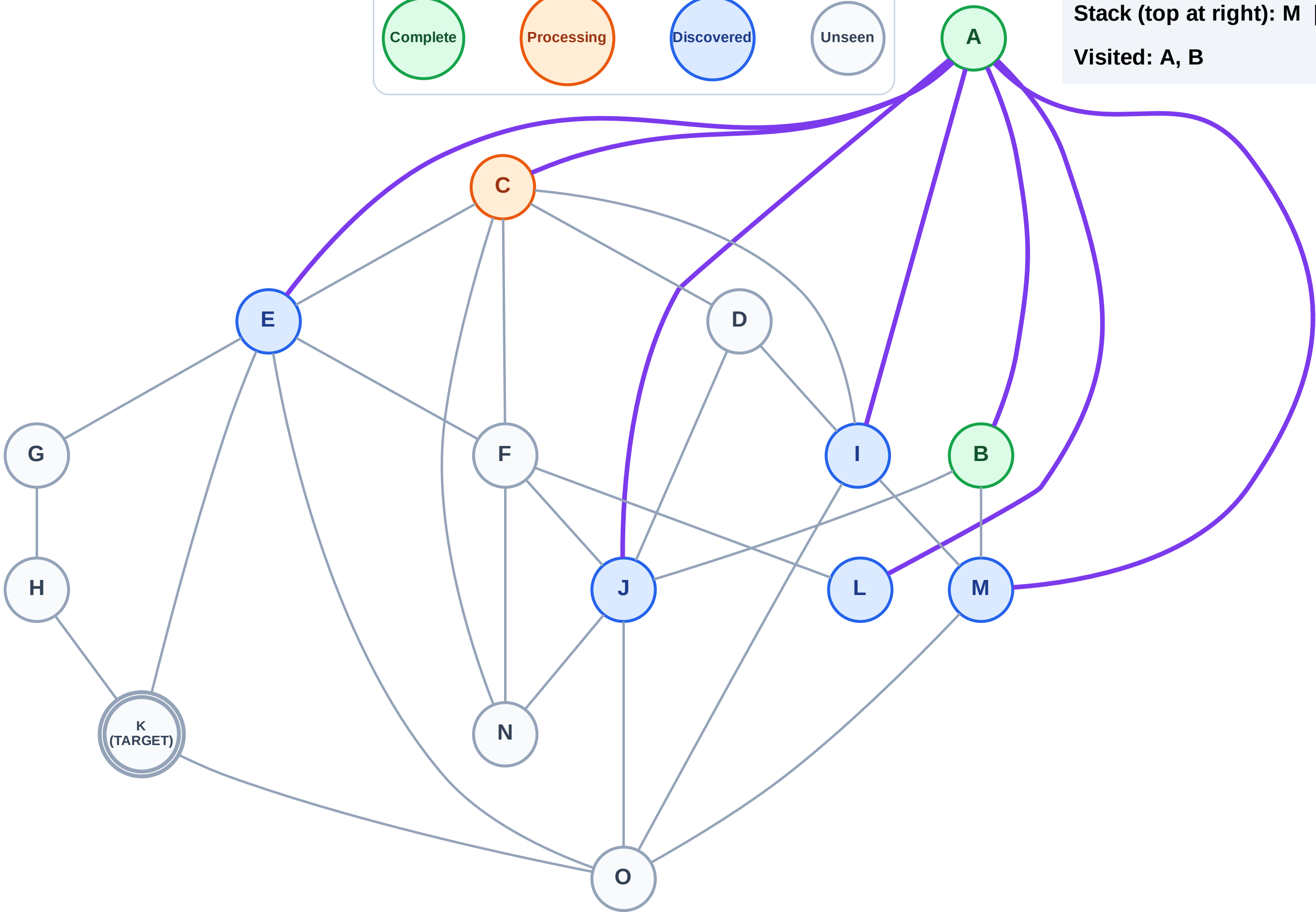
Step 6: Process node C

Legend



Stack (top at right): M | L | J | I | E

Visited: A, B



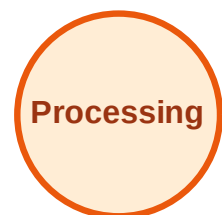
DFS Traversal

Step 7: Discover N, F, D from C

Legend



Complete



Processing



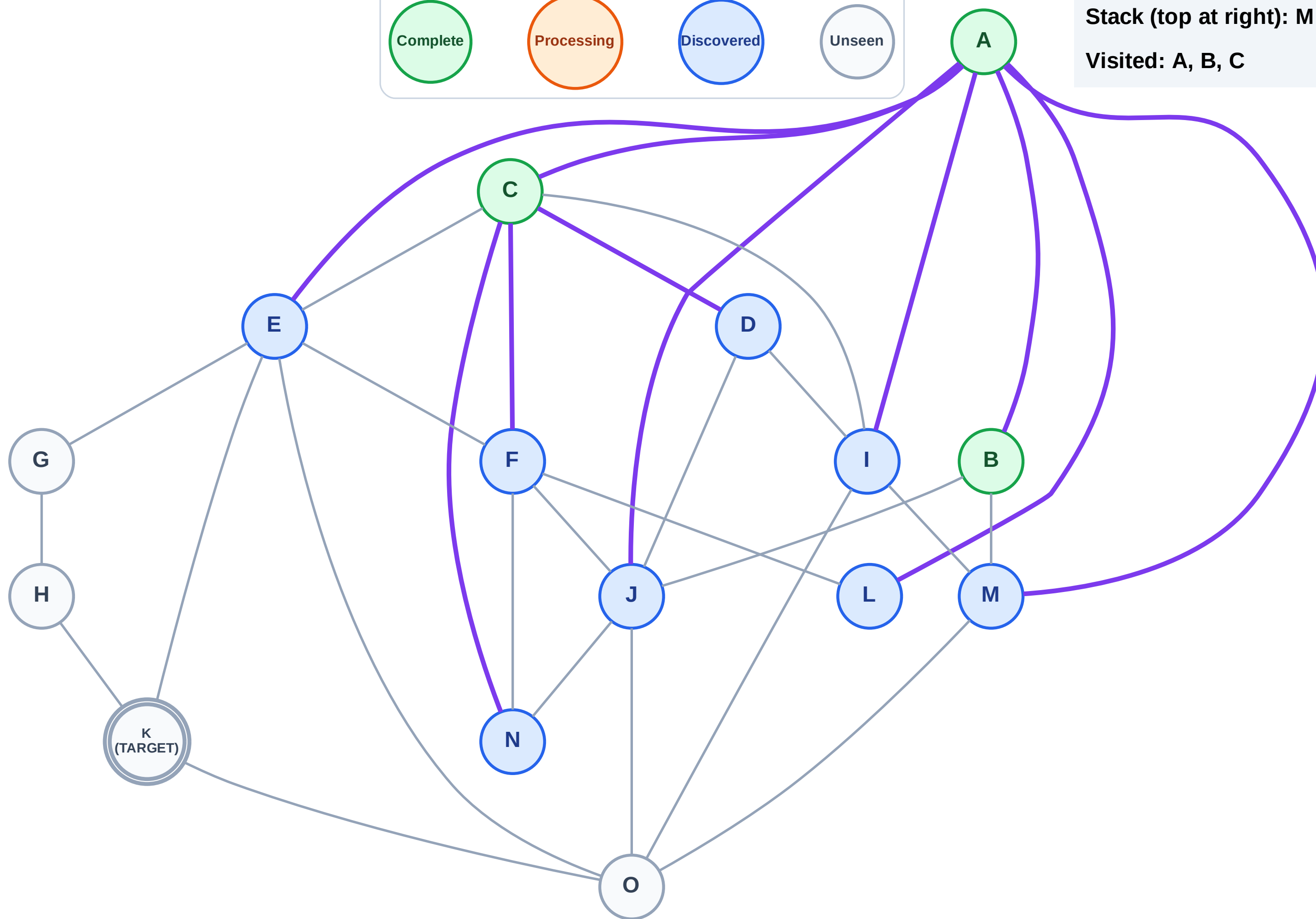
Discovered



Unseen

Stack (top at right): M | L | J | I | E | N | F | D

Visited: A, B, C



DFS Traversal

Step 8: Process node D

Legend

Complete

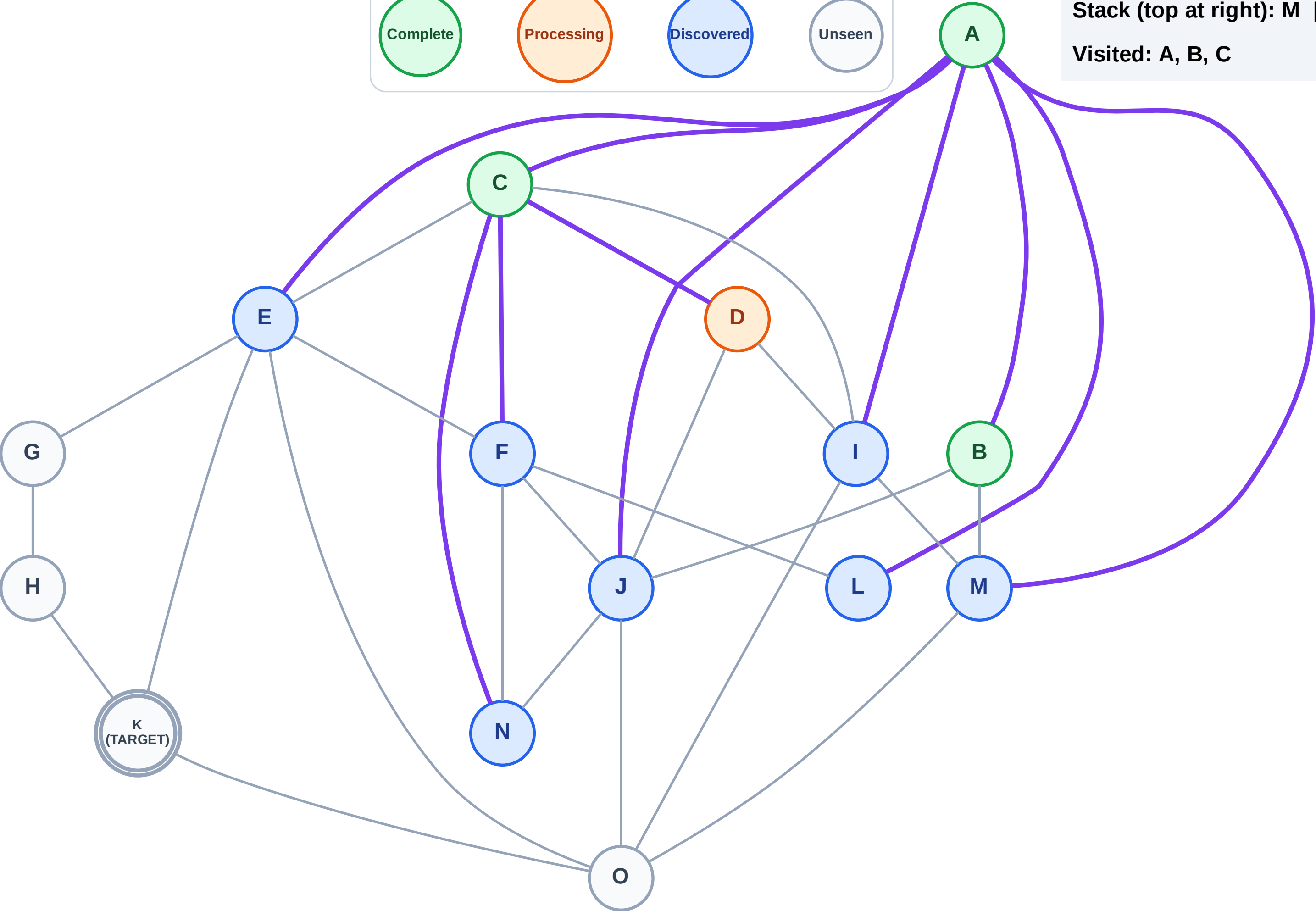
Processing

Discovered

Unseen

Stack (top at right): M | L | J | I | E | N | F

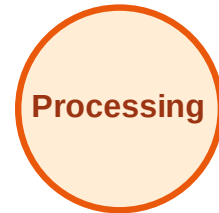
Visited: A, B, C



DFS Traversal

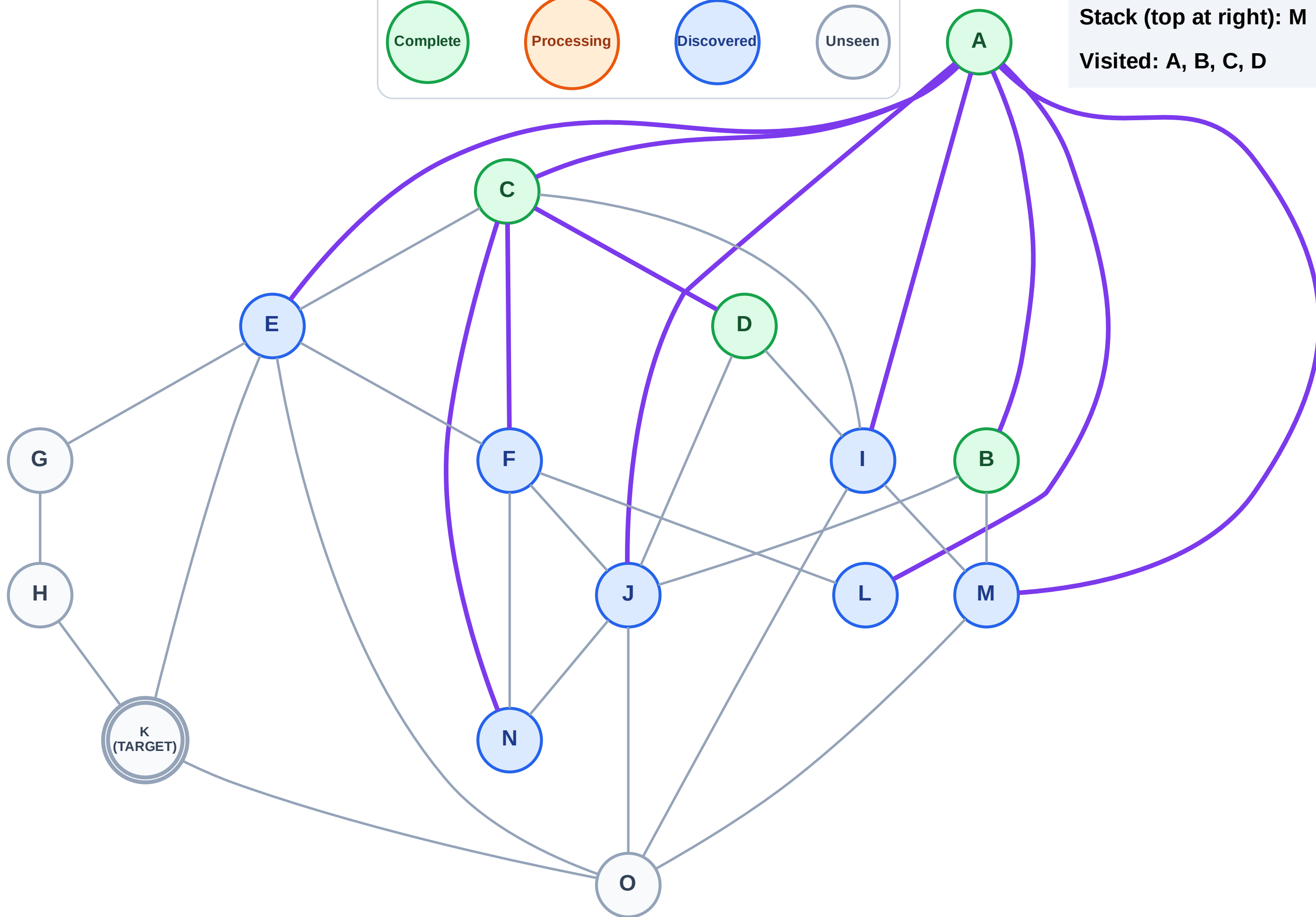
Step 9: No new neighbors from D

Legend



Stack (top at right): M | L | J | I | E | N | F

Visited: A, B, C, D



DFS Traversal

Step 10: Process node F

Legend

Complete

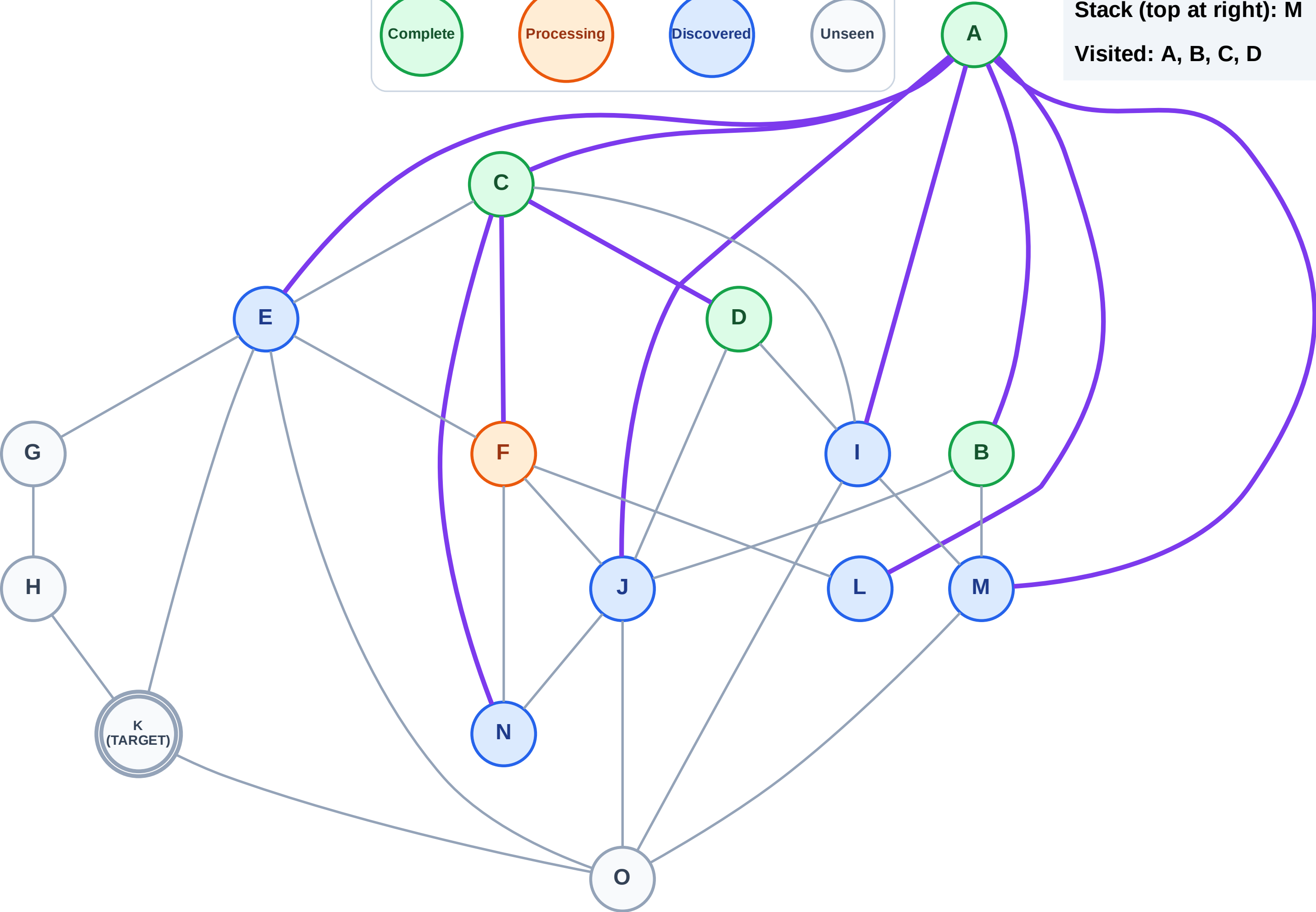
Processing

Discovered

Unseen

Stack (top at right): M | L | J | I | E | N

Visited: A, B, C, D



DFS Traversal

Step 11: No new neighbors from F

Legend

Complete

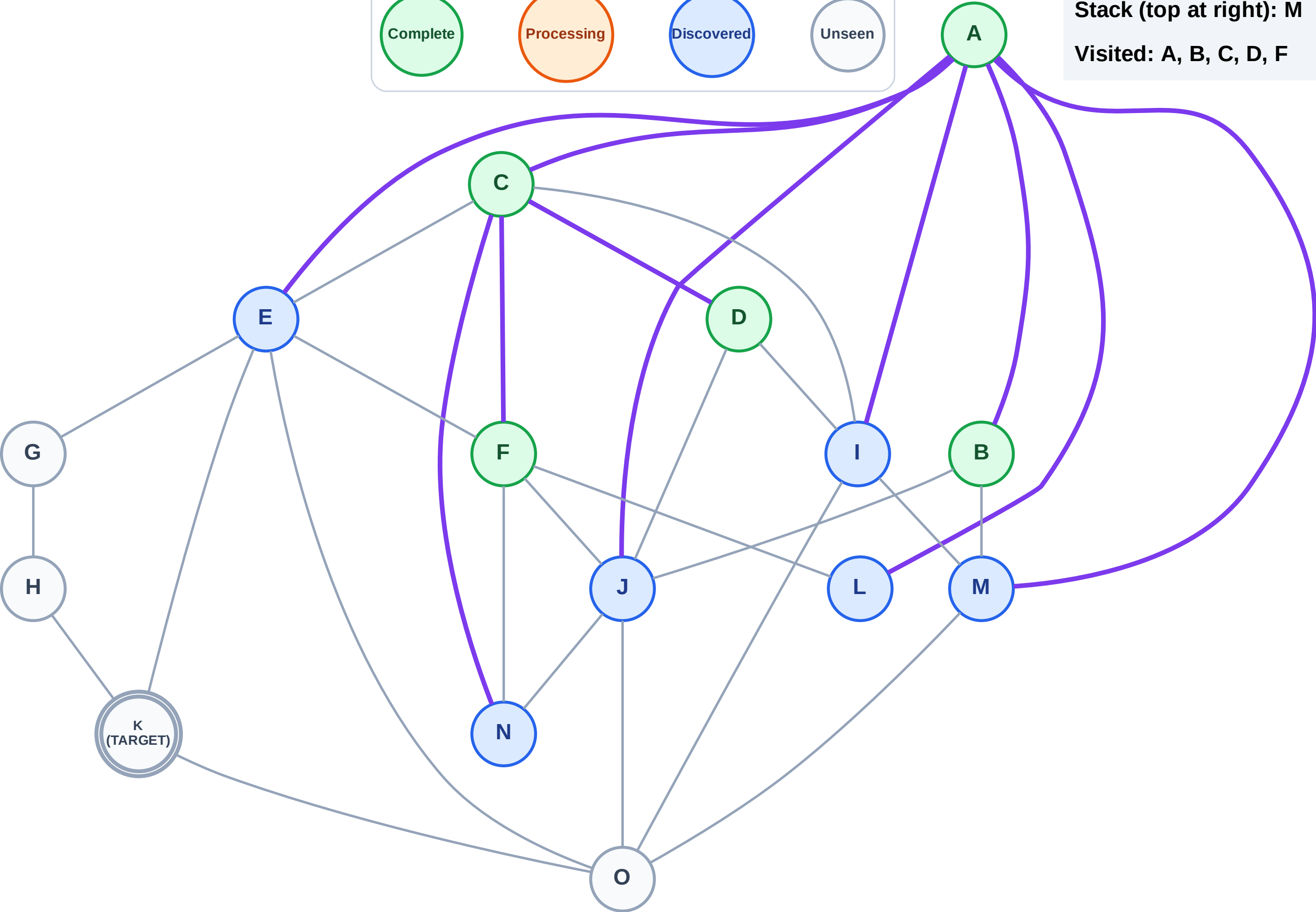
Processing

Discovered

Unseen

Stack (top at right): M | L | J | I | E | N

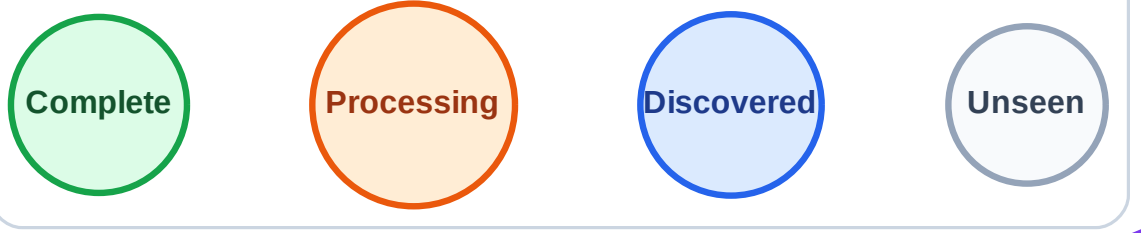
Visited: A, B, C, D, F



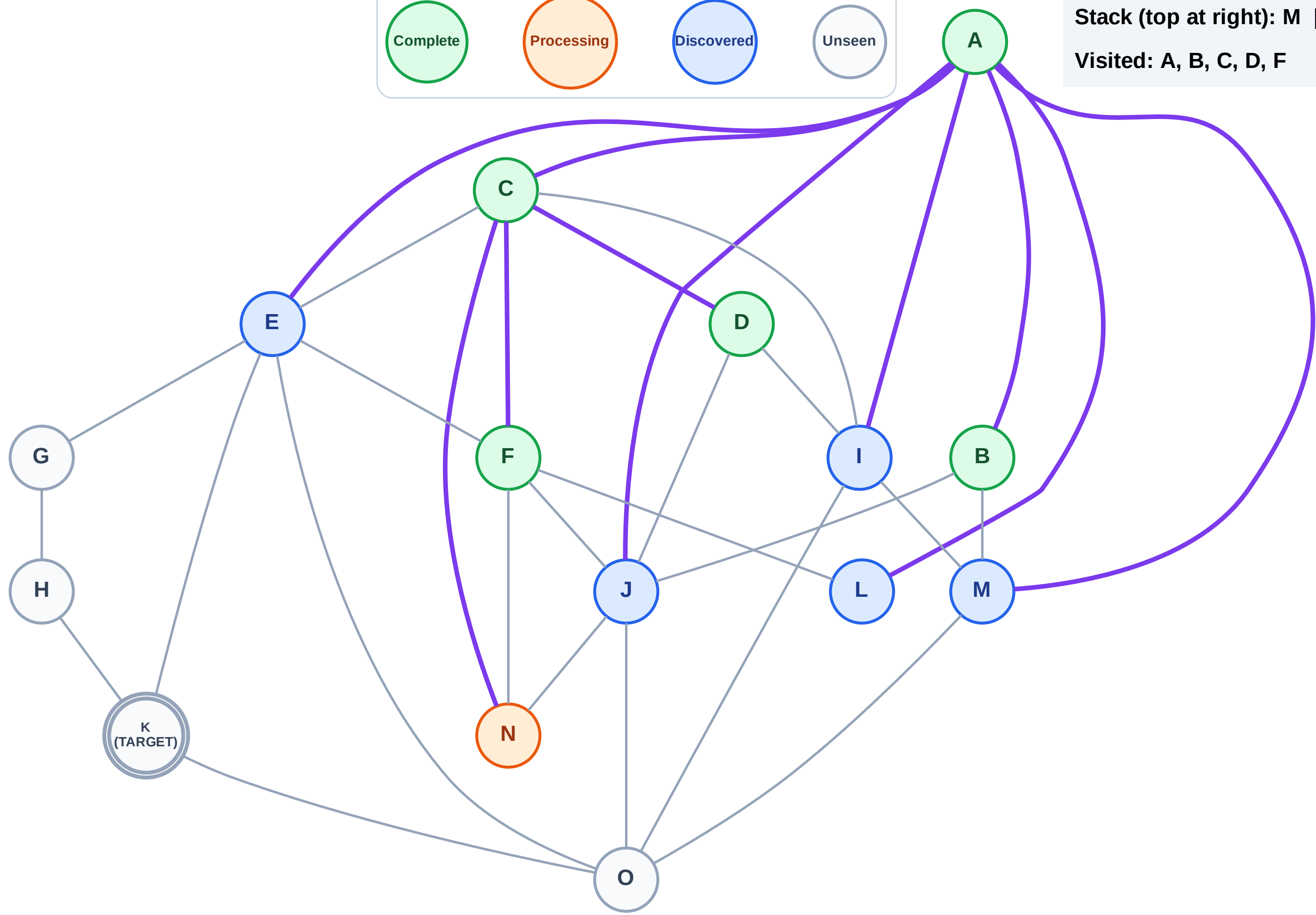
DFS Traversal

Step 12: Process node N

Legend



Stack (top at right): M | L | J | I | E
Visited: A, B, C, D, F



DFS Traversal

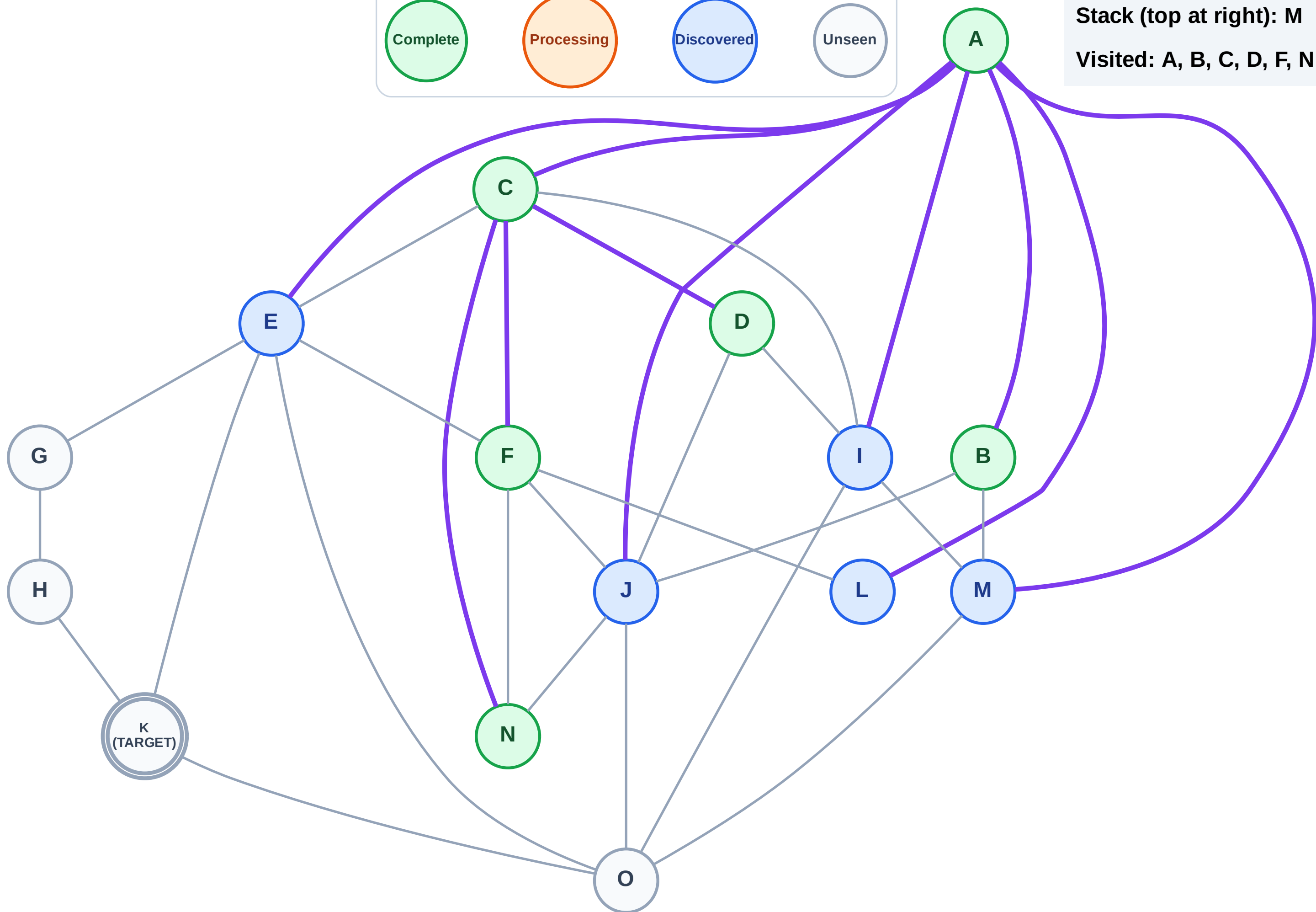
Step 13: No new neighbors from N

Legend



Stack (top at right): M | L | J | I | E

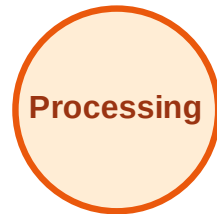
Visited: A, B, C, D, F, N



DFS Traversal

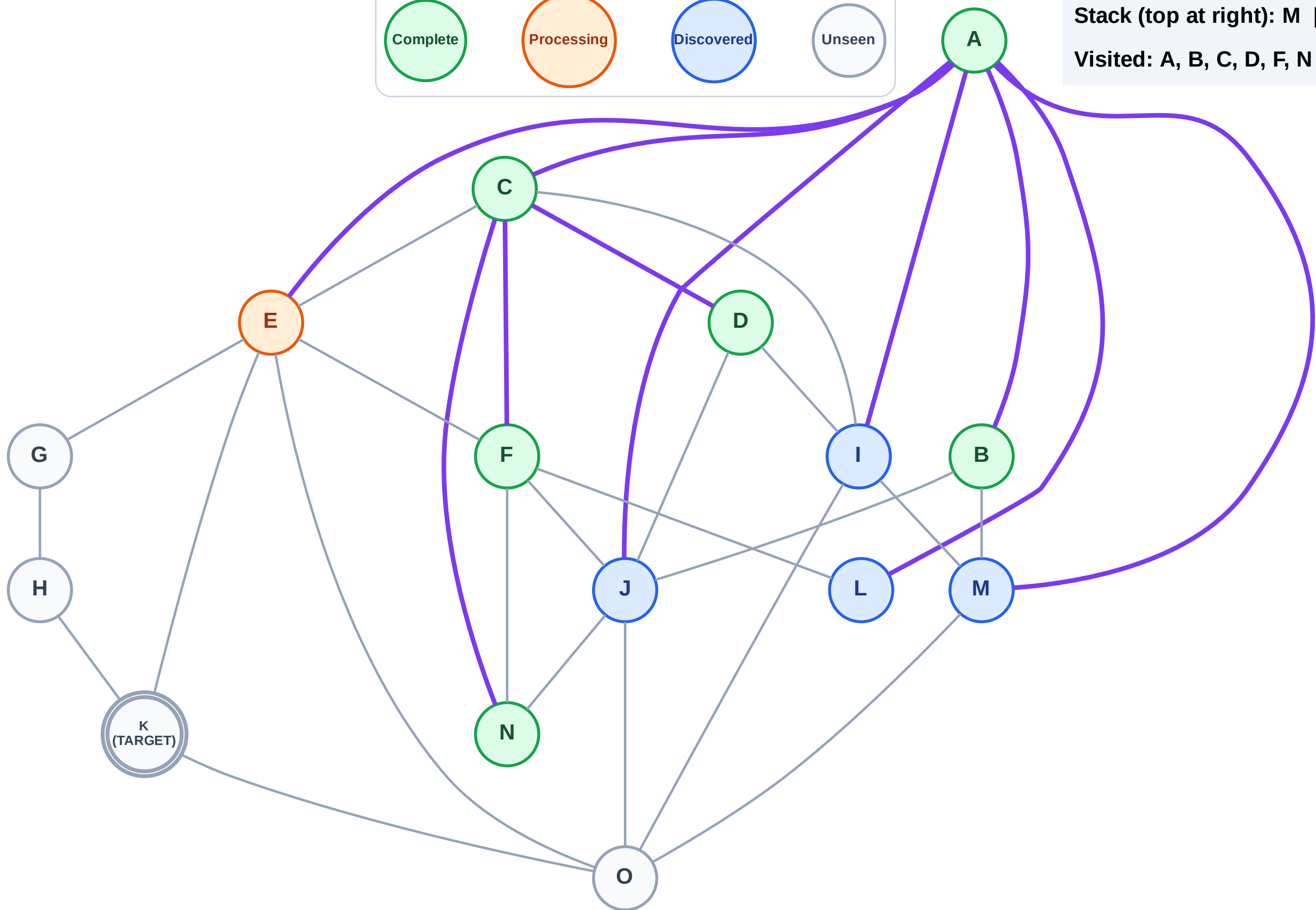
Step 14: Process node E

Legend



Stack (top at right): M | L | J | I

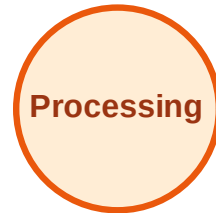
Visited: A, B, C, D, F, N



DFS Traversal

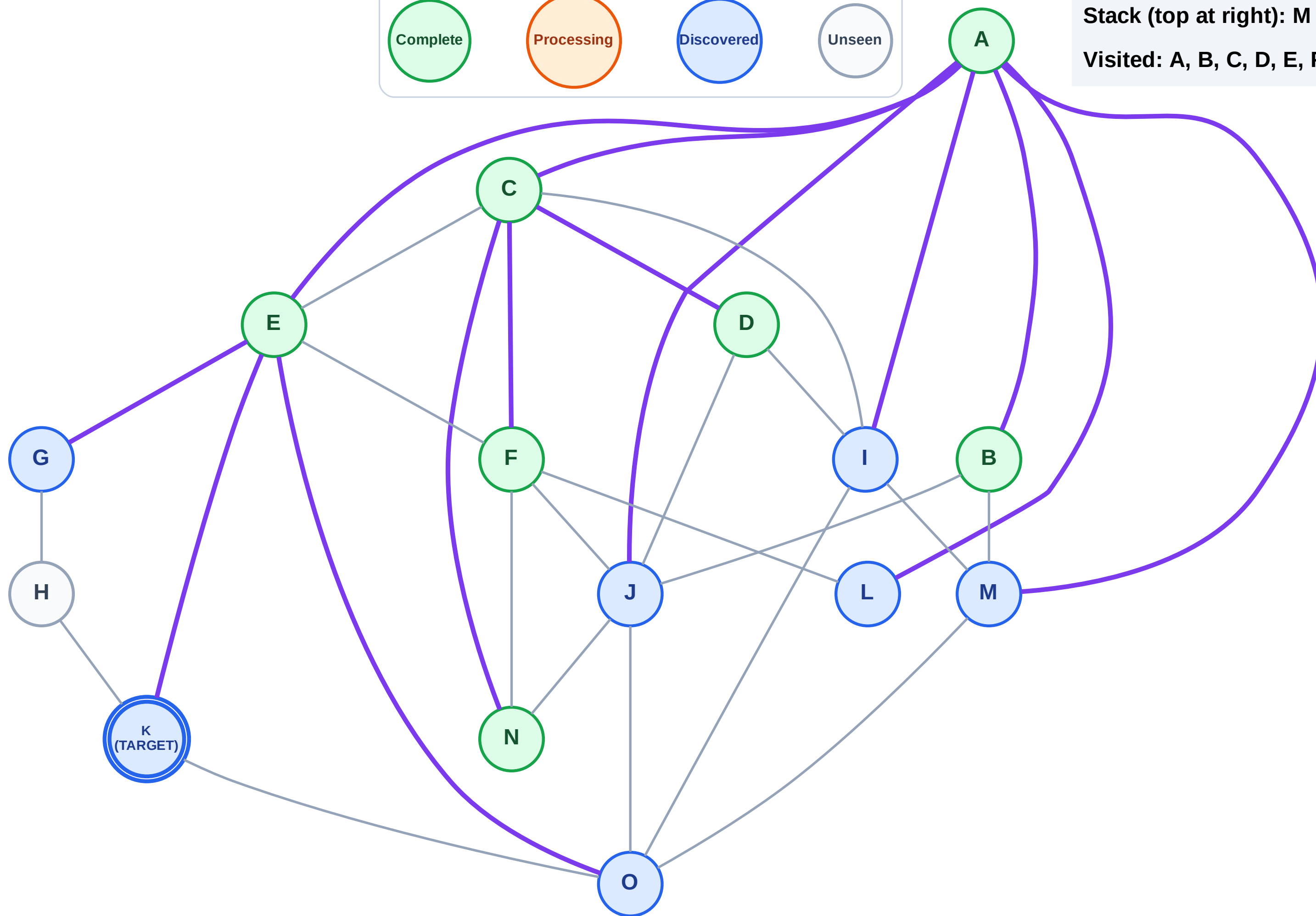
Step 15: Discover O, K, G from E

Legend



Stack (top at right): M | L | J | I | O | K | G

Visited: A, B, C, D, E, F, N



DFS Traversal

Step 16: Process node G

Legend

Complete

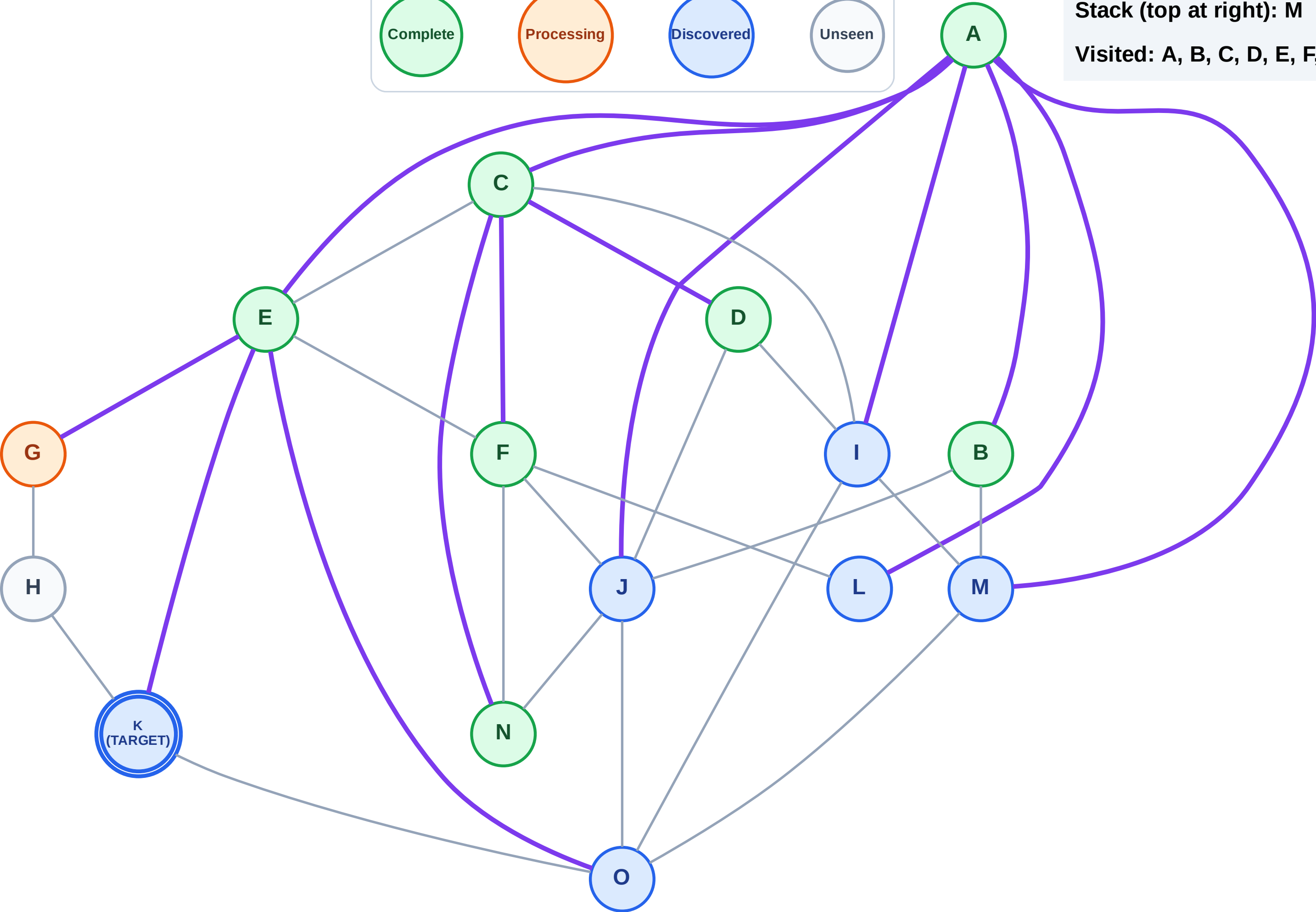
Processing

Discovered

Unseen

Stack (top at right): M | L | J | I | O | K

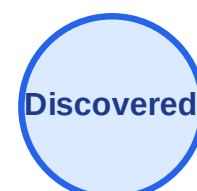
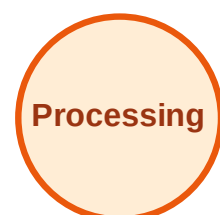
Visited: A, B, C, D, E, F, N



DFS Traversal

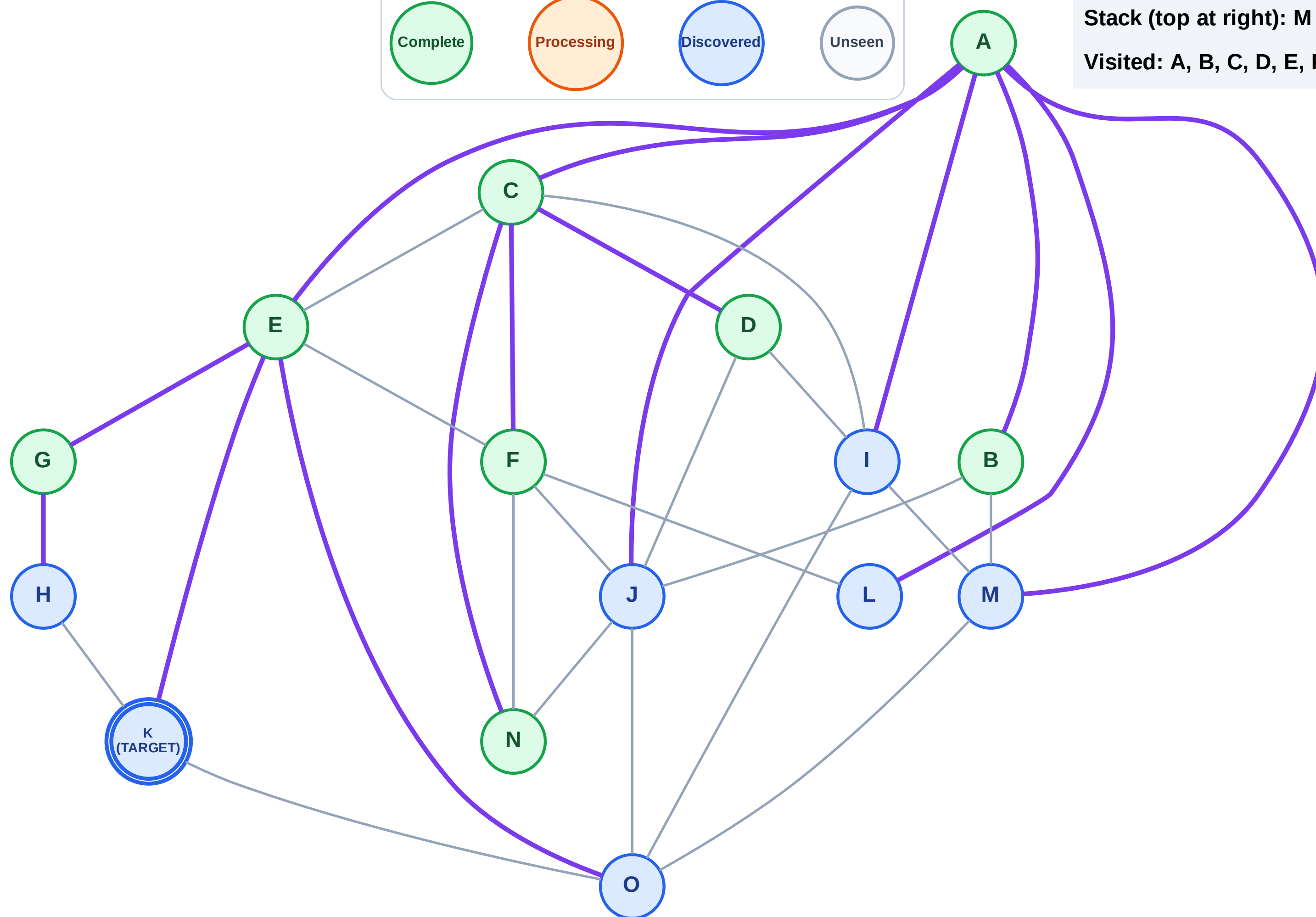
Step 17: Discover H from G

Legend



Stack (top at right): M | L | J | I | O | K | H

Visited: A, B, C, D, E, F, G, N



DFS Traversal

Step 18: Process node H

Legend

Complete

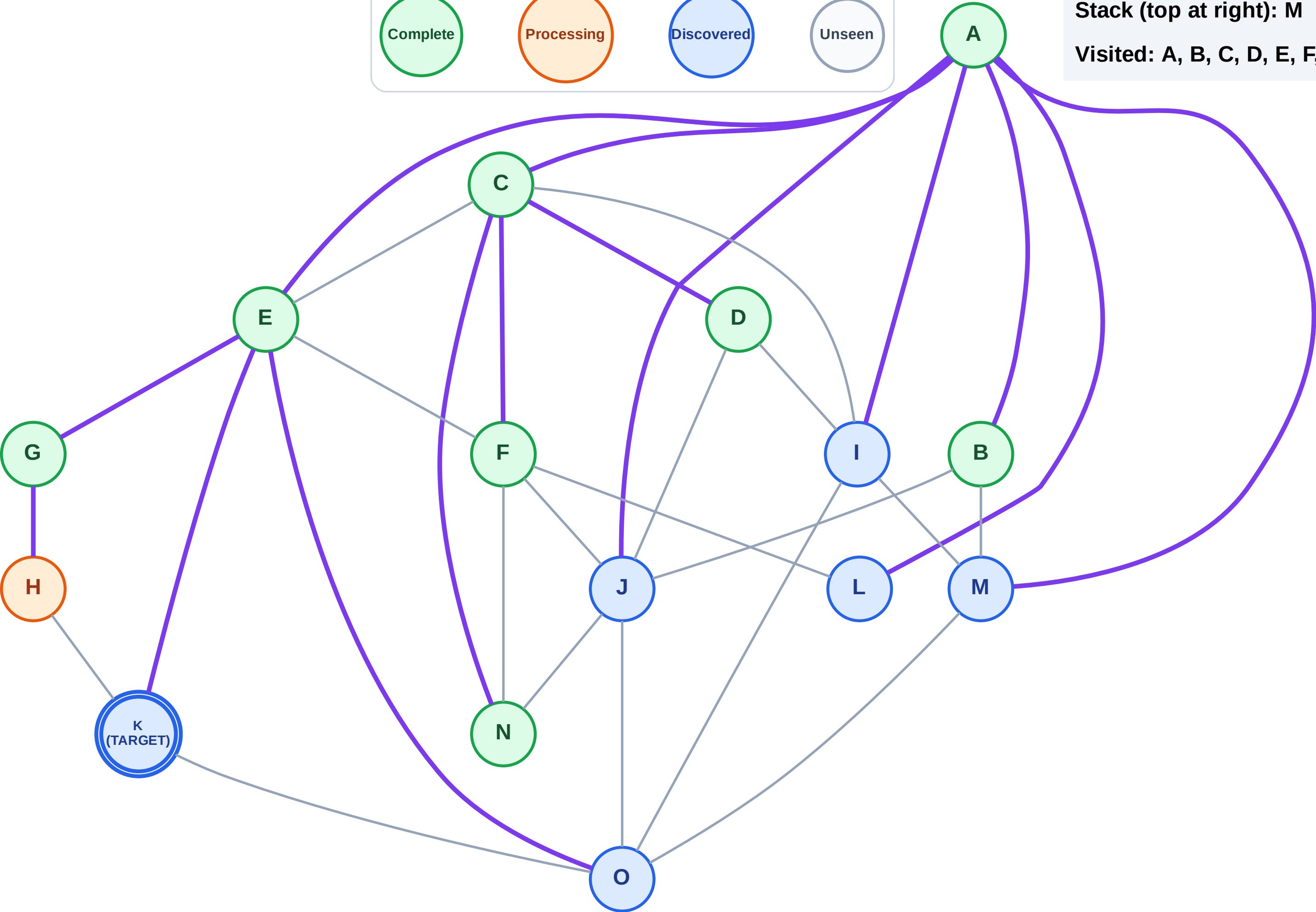
Processing

Discovered

Unseen

Stack (top at right): M | L | J | I | O | K

Visited: A, B, C, D, E, F, G, N



DFS Traversal

Step 19: No new neighbors from H

Legend

Complete

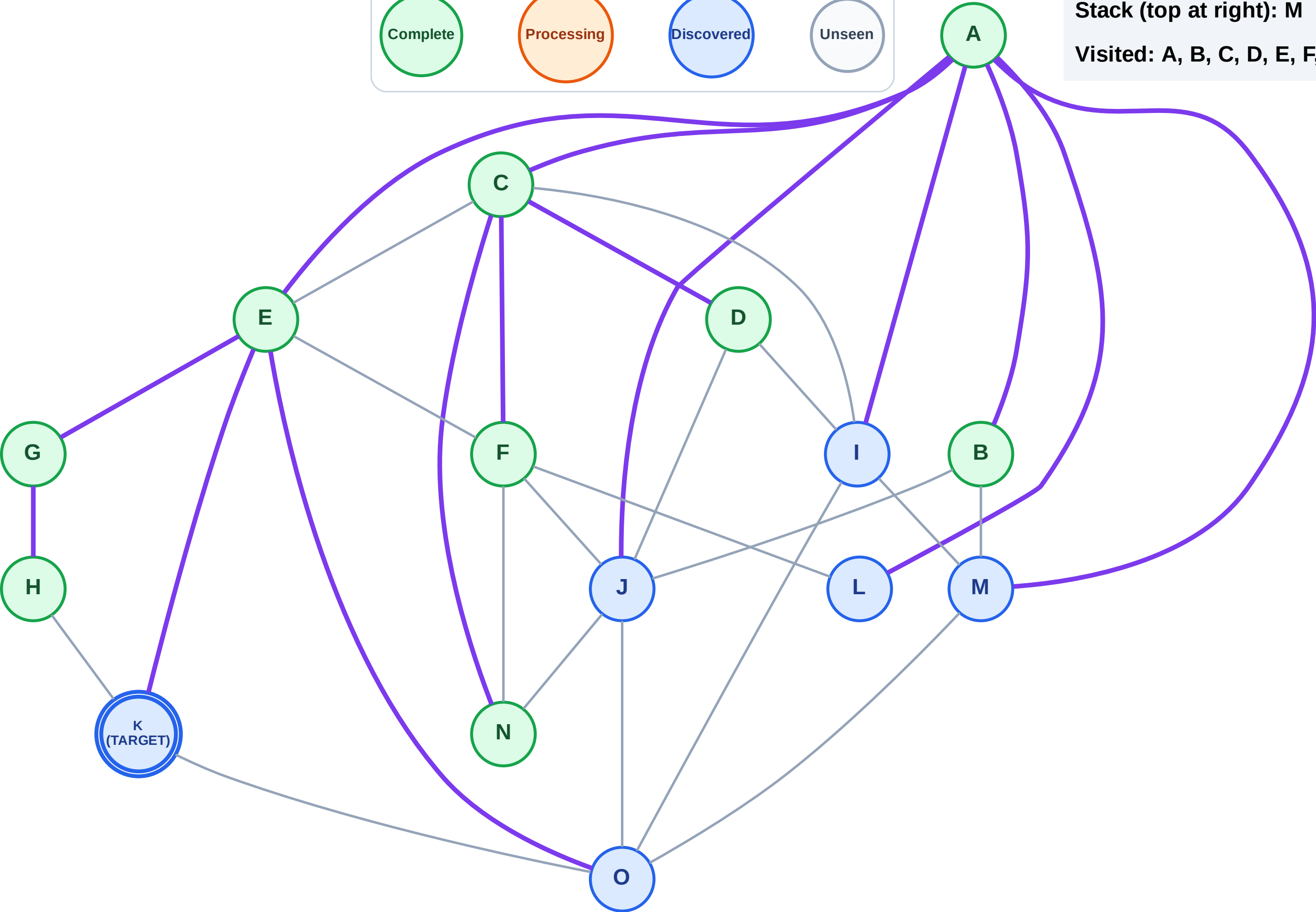
Processing

Discovered

Unseen

Stack (top at right): M | L | J | I | O | K

Visited: A, B, C, D, E, F, G, H, N



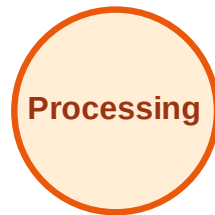
DFS Traversal

Step 20: Process node K

Legend



Complete



Processing



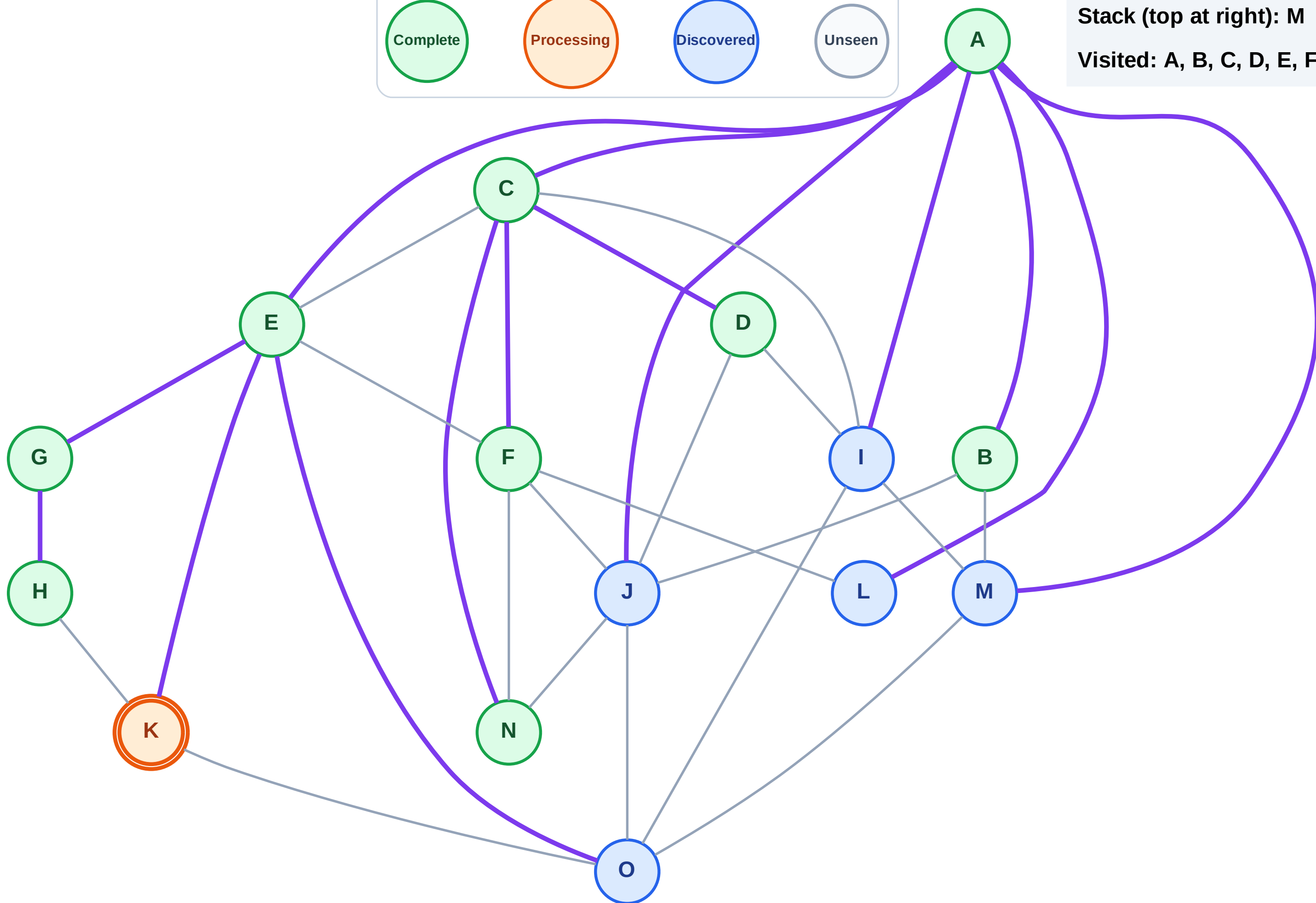
Discovered



Unseen

Stack (top at right): M | L | J | I | O

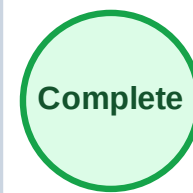
Visited: A, B, C, D, E, F, G, H, N



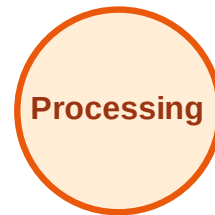
DFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → E → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): M | L | J | I | O

Visited: A, B, C, D, E, F, G, H, K, N

